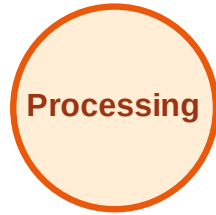


BFS Traversal

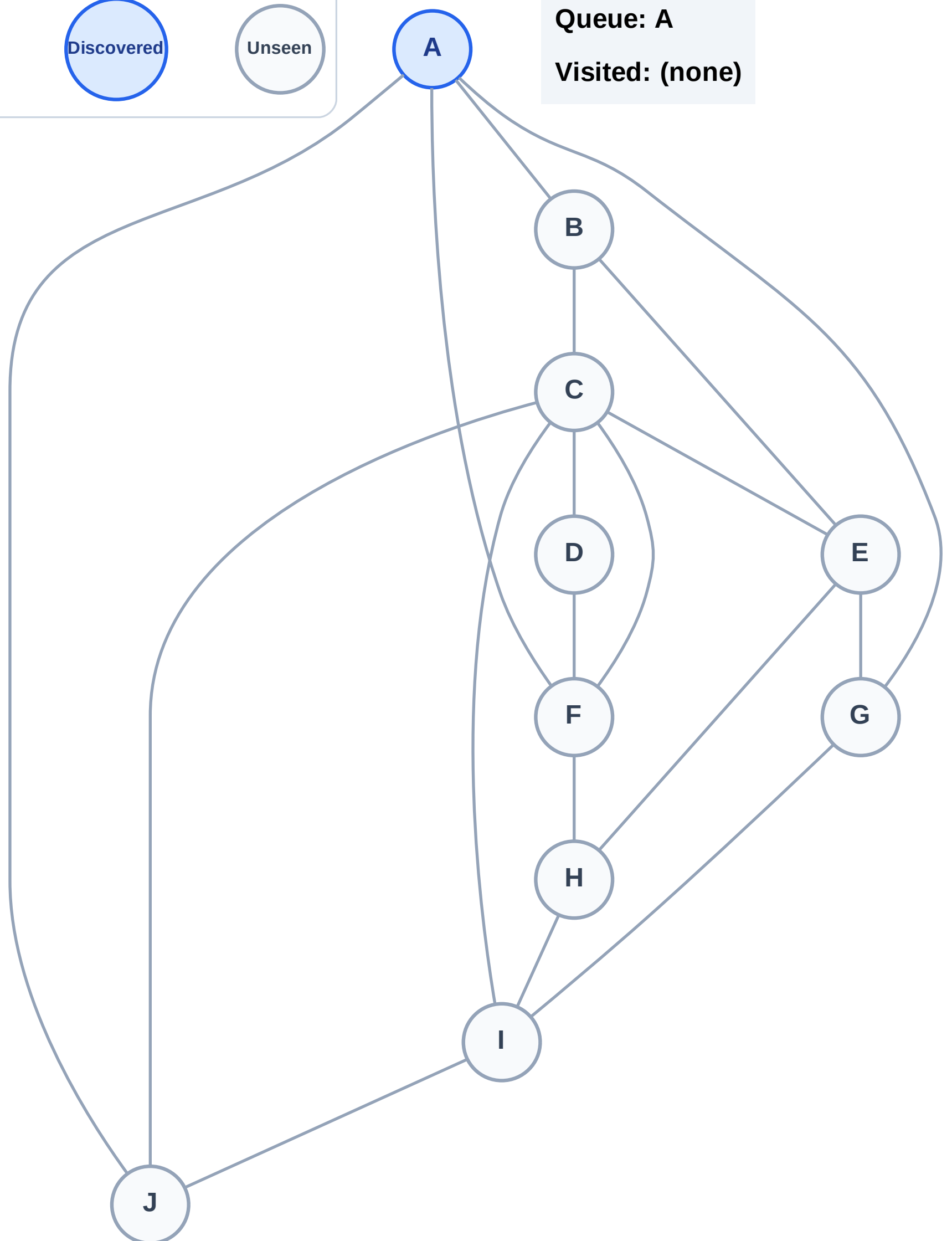
Step 1: Start at A

Legend



Queue: A

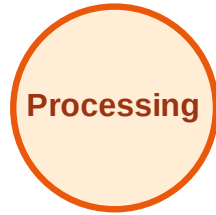
Visited: (none)



BFS Traversal

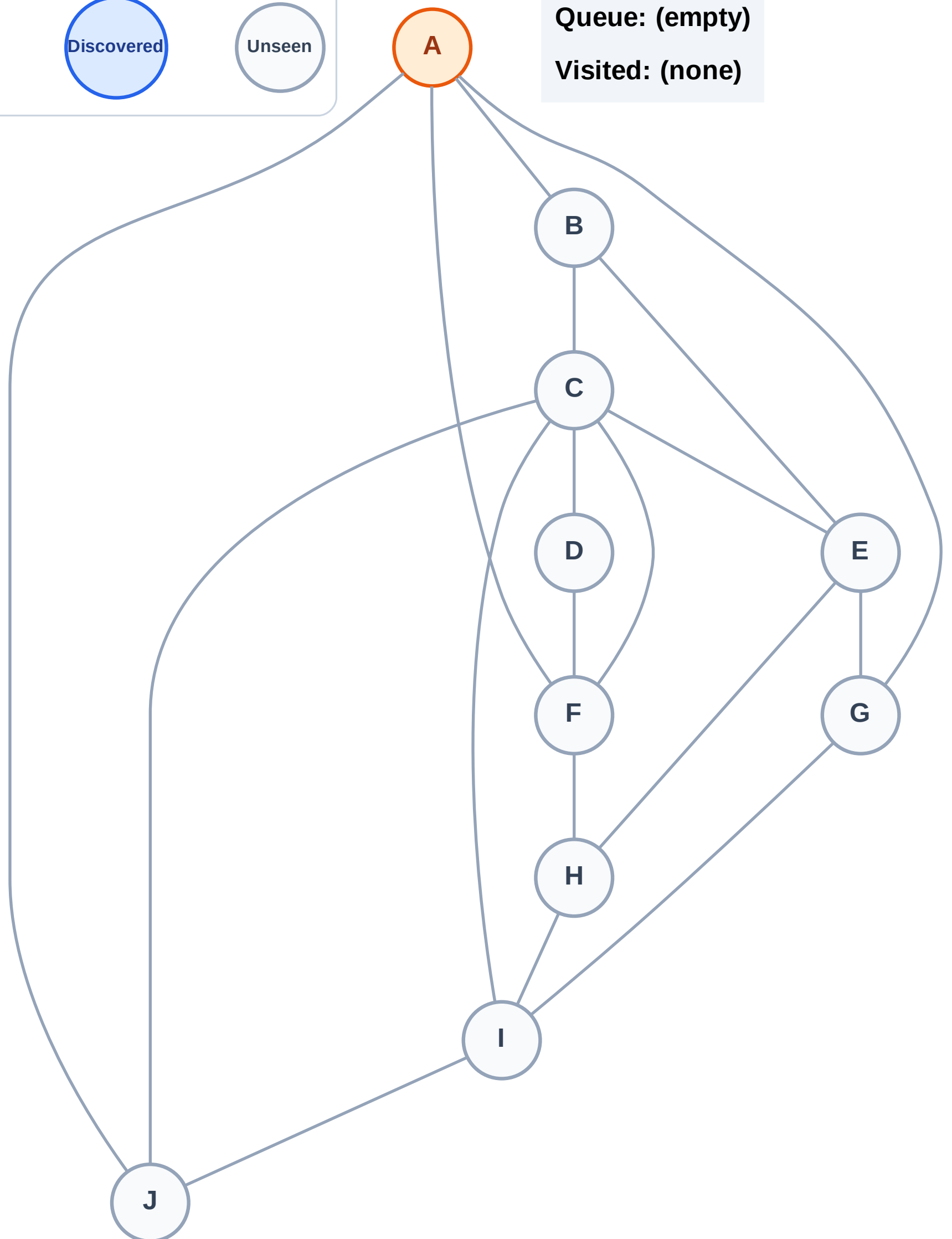
Step 2: Process node A

Legend



Queue: (empty)

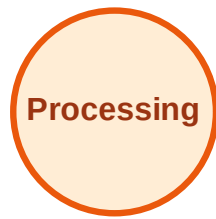
Visited: (none)



BFS Traversal

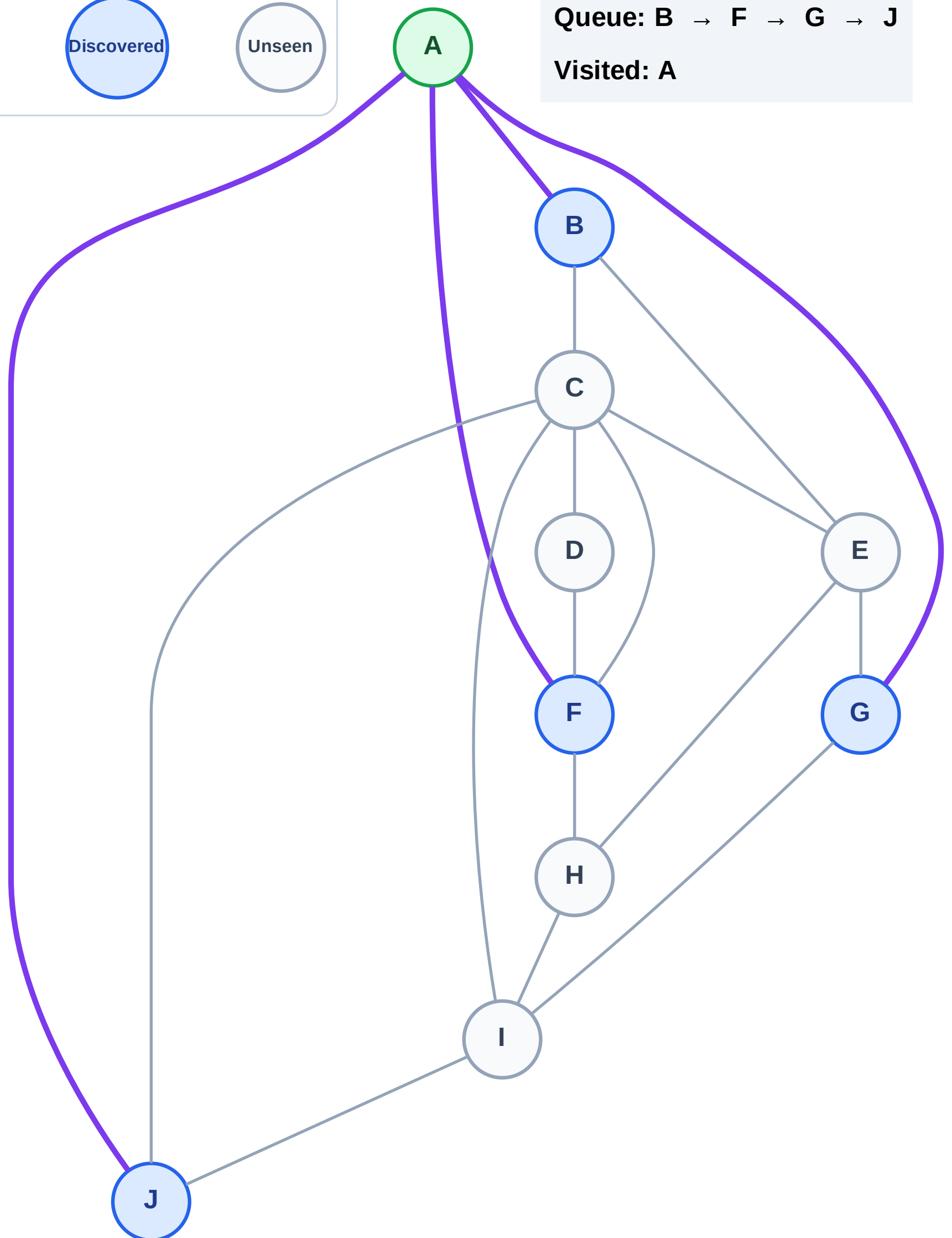
Step 3: Discover B, F, G, J from A

Legend



Queue: B → F → G → J

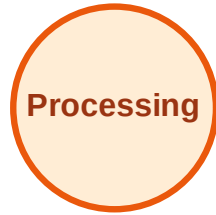
Visited: A



BFS Traversal

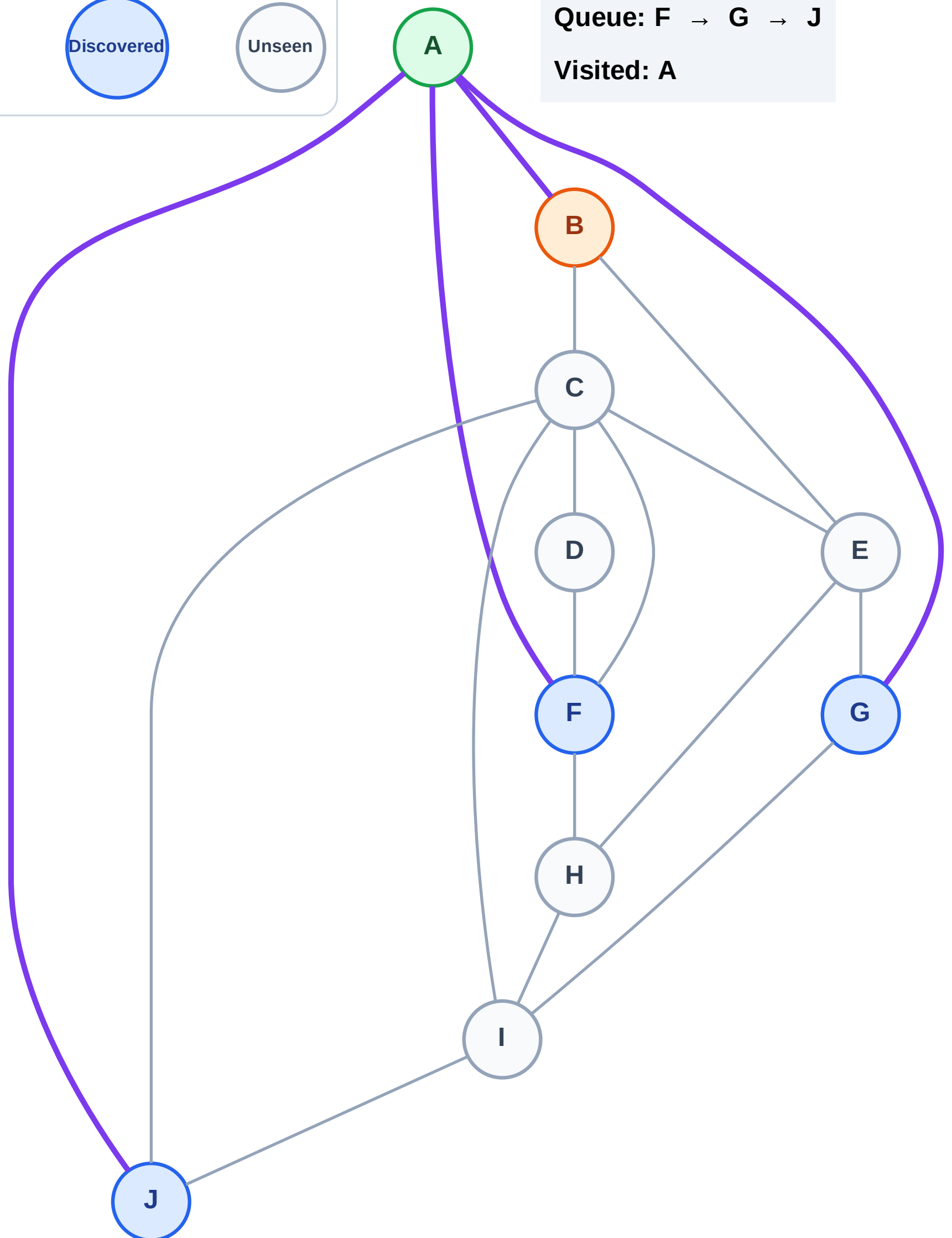
Step 4: Process node B

Legend



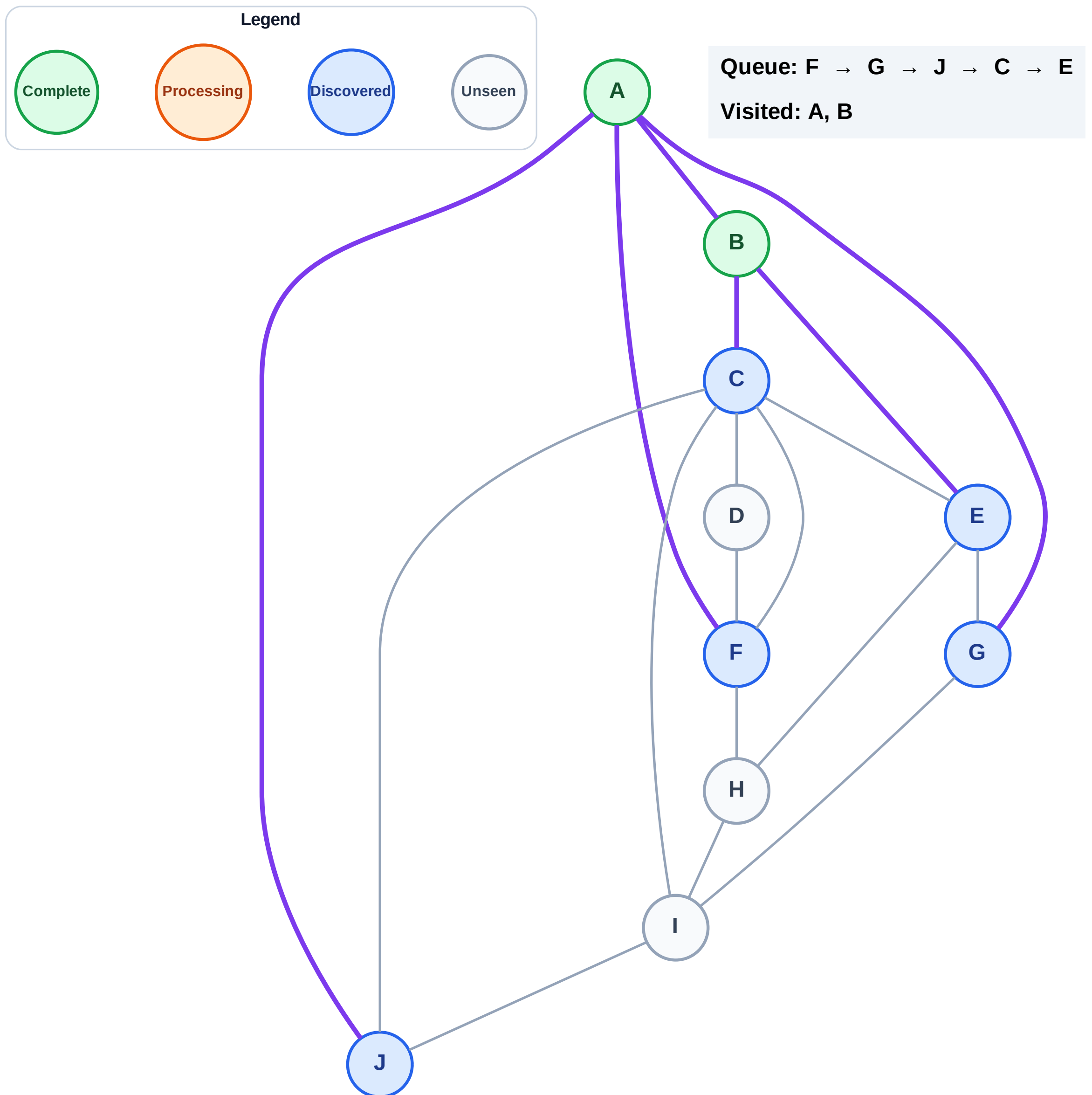
Queue: F → G → J

Visited: A



Step 5: Discover C, E from B

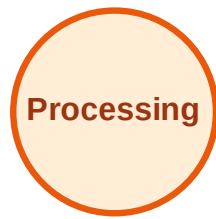
Step 5: Discover C, E from B



BFS Traversal

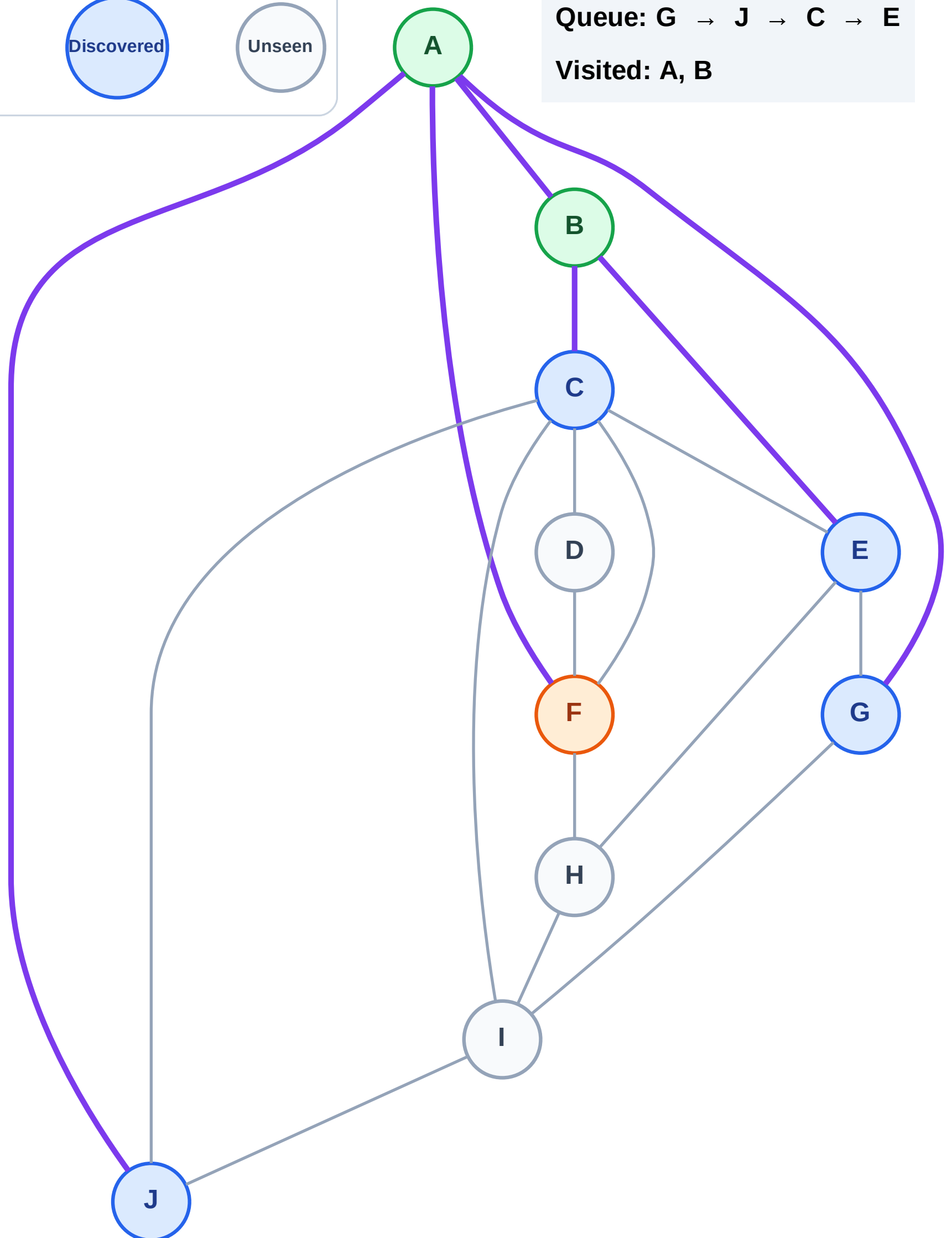
Step 6: Process node F

Legend



Queue: G → J → C → E

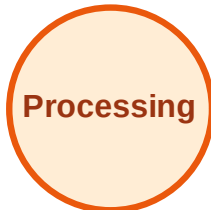
Visited: A, B



BFS Traversal

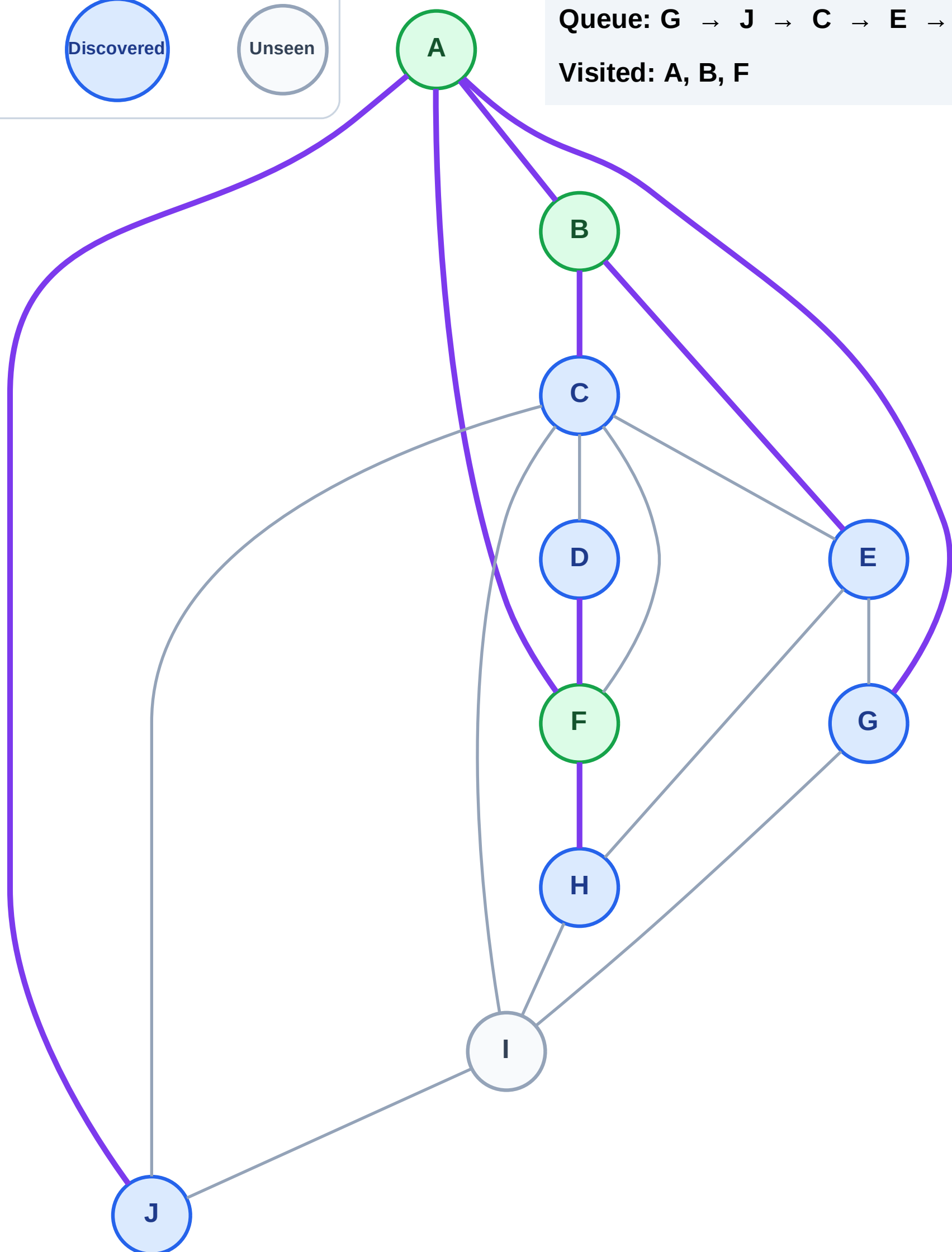
Step 7: Discover D, H from F

Legend



Queue: G → J → C → E → D → H

Visited: A, B, F



BFS Traversal

Step 8: Process node G

Legend

Complete

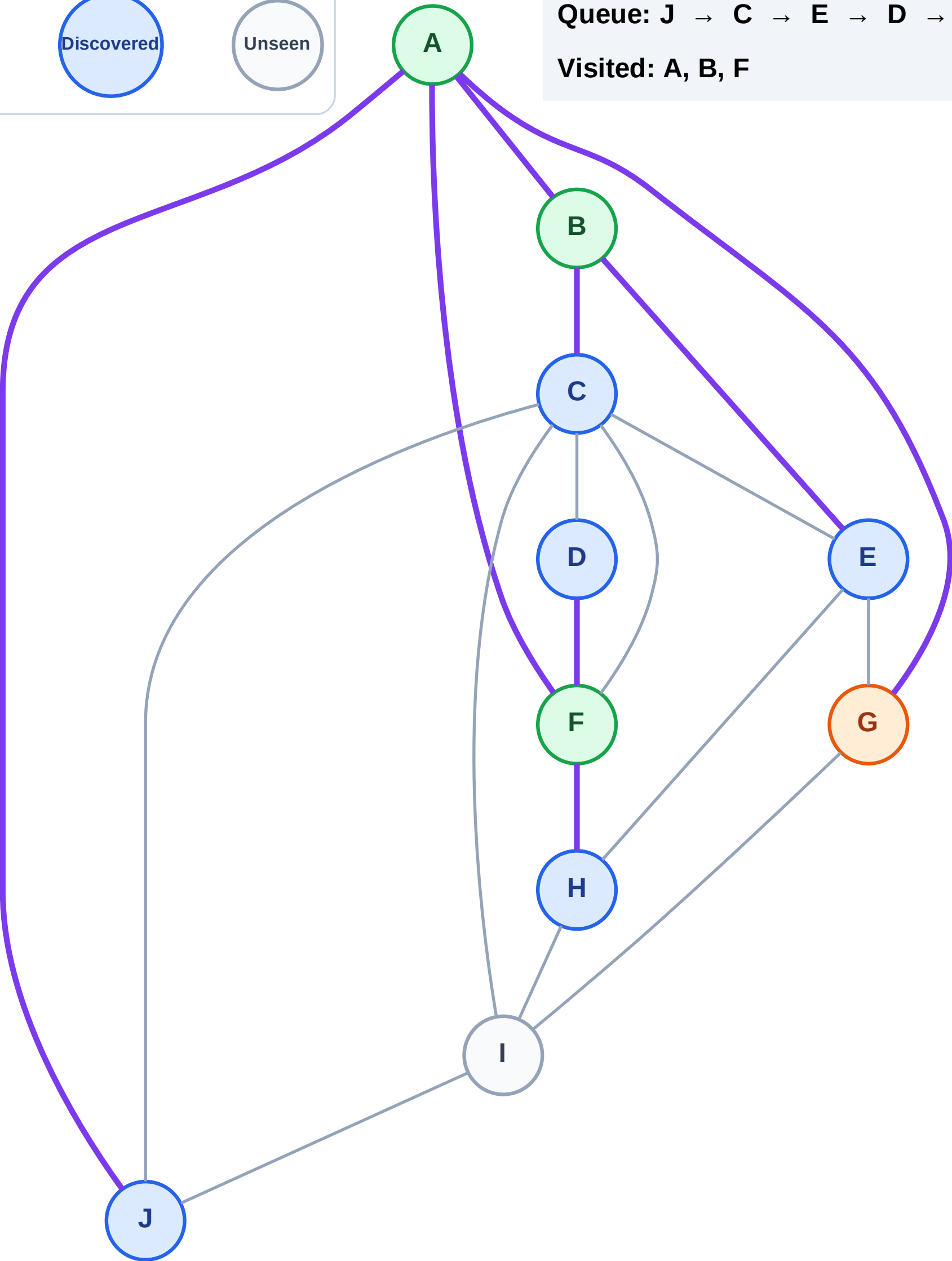
Processing

Discovered

Unseen

Queue: J → C → E → D → H

Visited: A, B, F



BFS Traversal

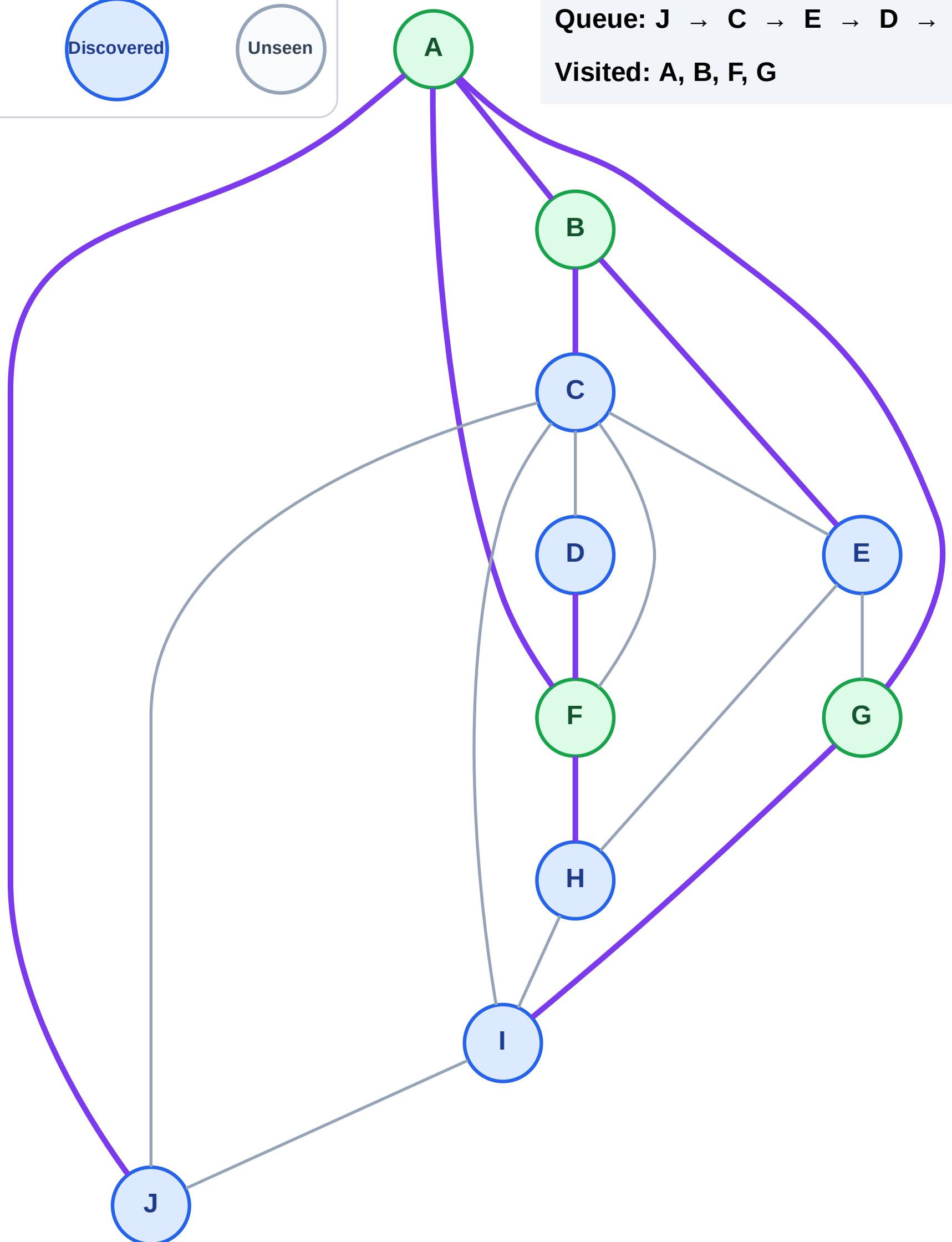
Step 9: Discover I from G

Legend



Queue: J → C → E → D → H → I

Visited: A, B, F, G



BFS Traversal

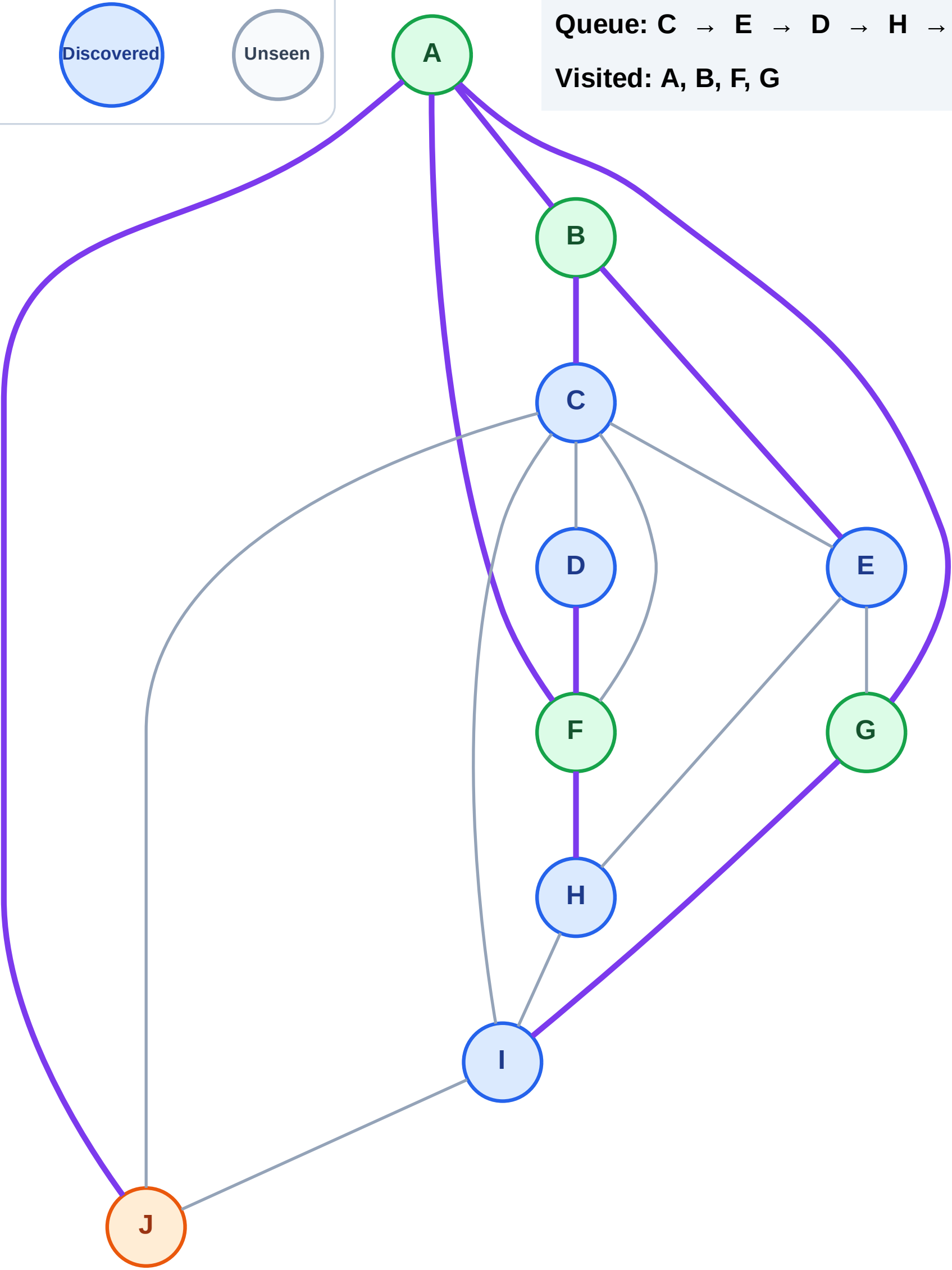
Step 10: Process node J

Legend



Queue: C → E → D → H → I

Visited: A, B, F, G



BFS Traversal

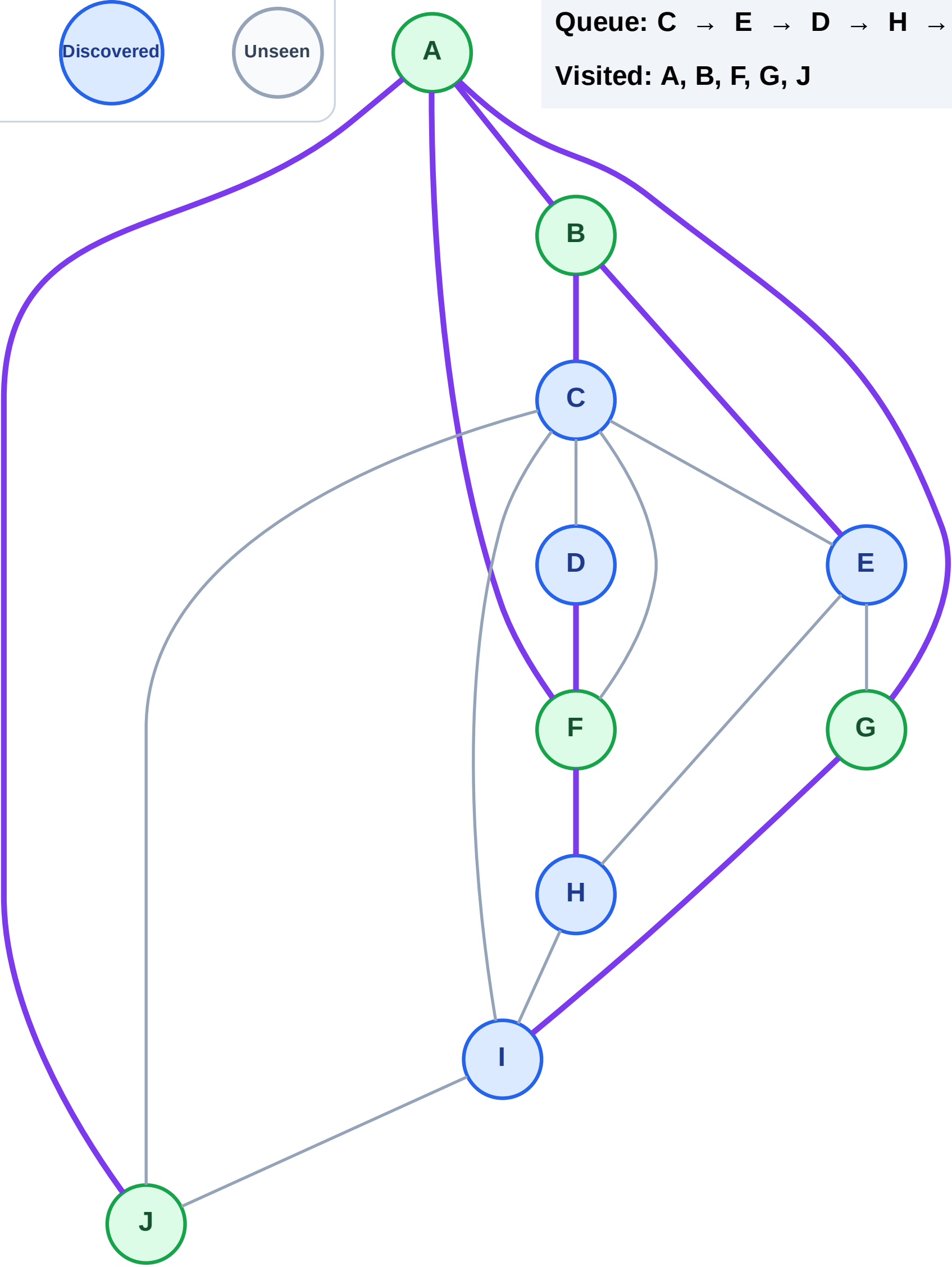
Step 11: No new neighbors from J

Legend



Queue: C → E → D → H → I

Visited: A, B, F, G, J



BFS Traversal

Step 12: Process node C

Legend

Complete

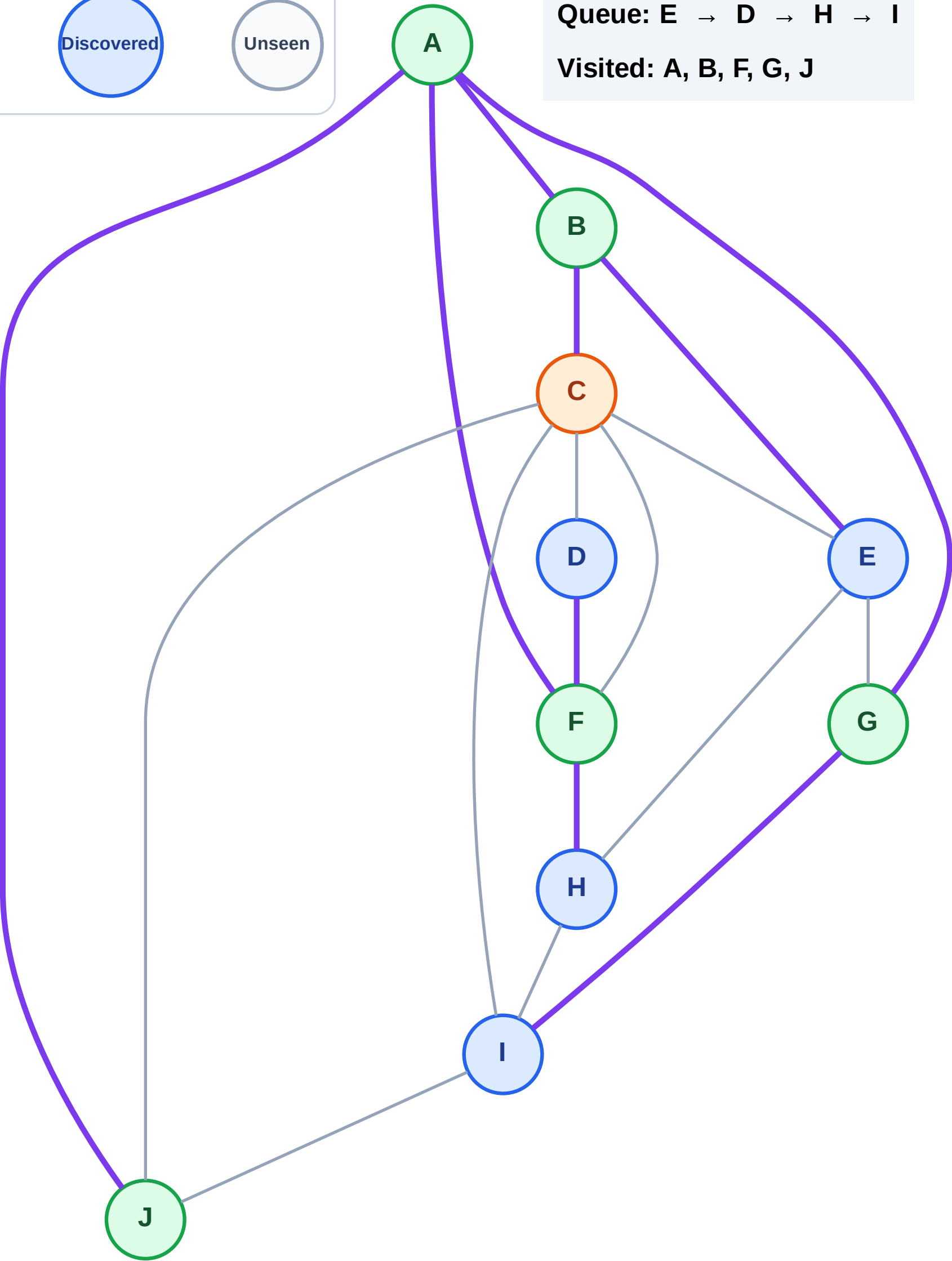
Processing

Discovered

Unseen

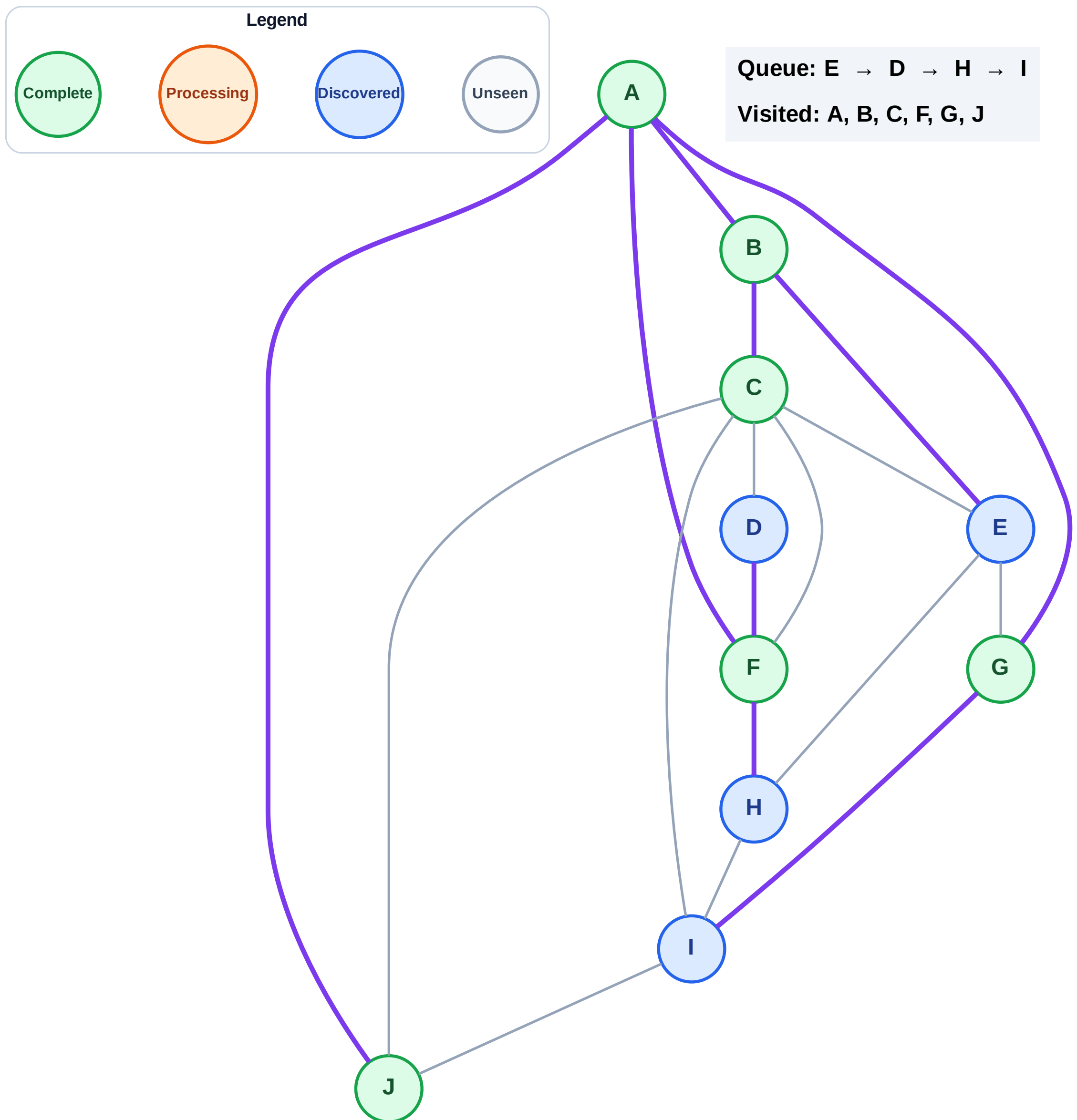
Queue: E → D → H → I

Visited: A, B, F, G, J



Step 13: No new neighbors from C

Step 13: No new neighbors from C



BFS Traversal

Step 14: Process node E

Legend

Complete

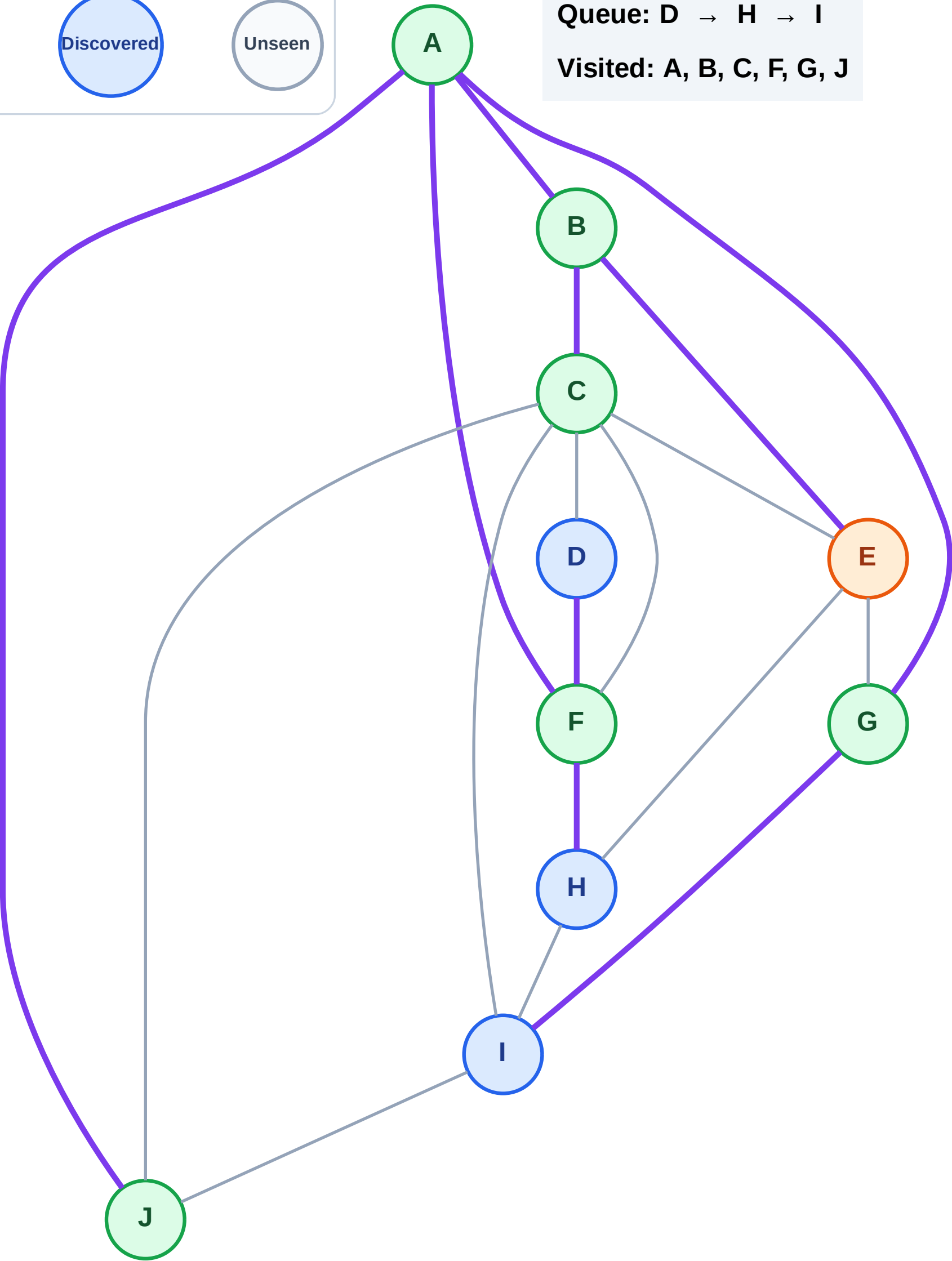
Processing

Discovered

Unseen

Queue: D → H → I

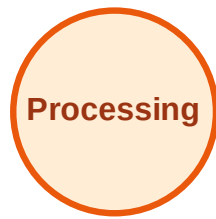
Visited: A, B, C, F, G, J



BFS Traversal

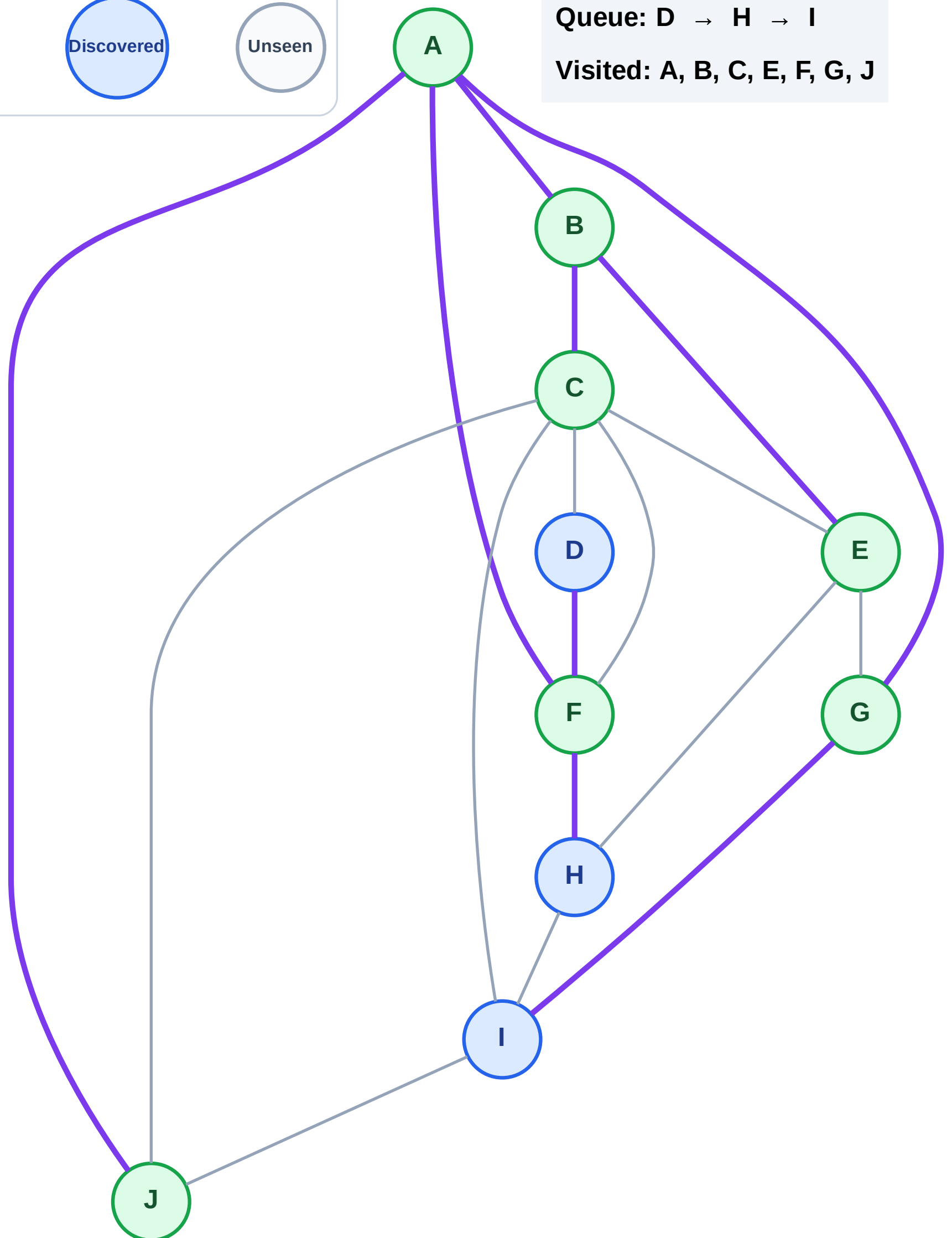
Step 15: No new neighbors from E

Legend



Queue: D → H → I

Visited: A, B, C, E, F, G, J



BFS Traversal

Step 16: Process node D

Legend

Complete

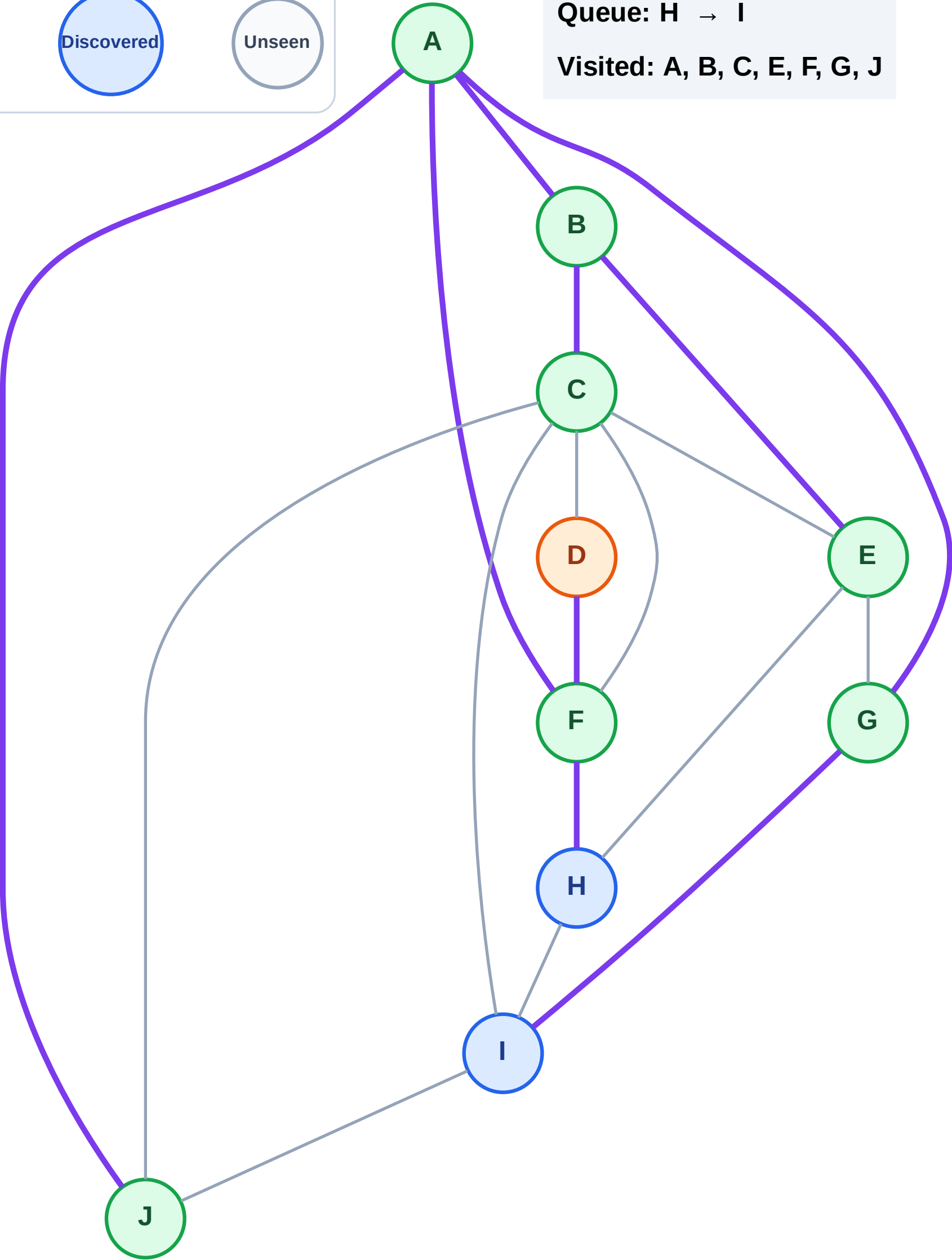
Processing

Discovered

Unseen

Queue: H → I

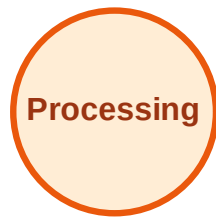
Visited: A, B, C, E, F, G, J



BFS Traversal

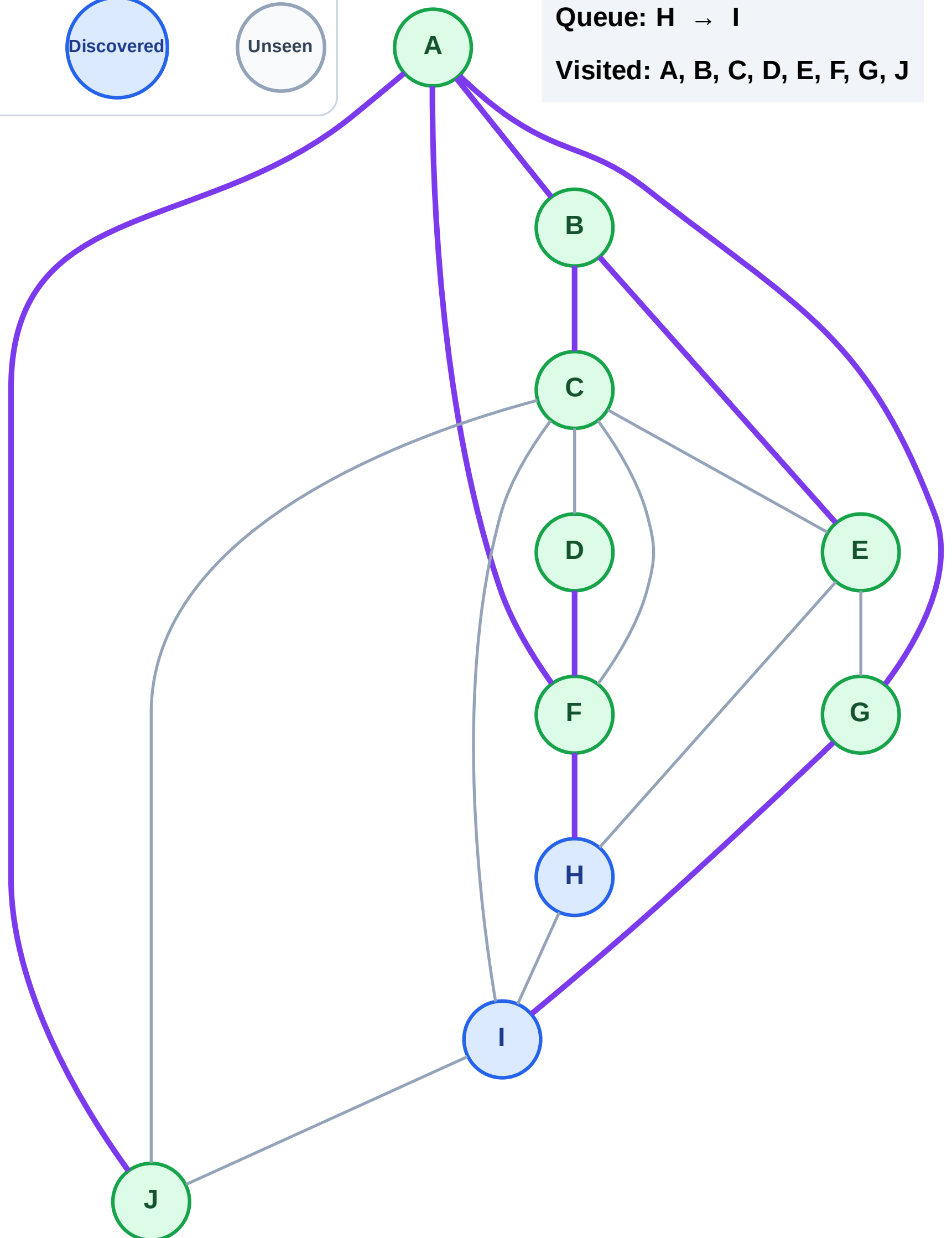
Step 17: No new neighbors from D

Legend



Queue: H → I

Visited: A, B, C, D, E, F, G, J



BFS Traversal

Step 18: Process node H

Legend

Complete

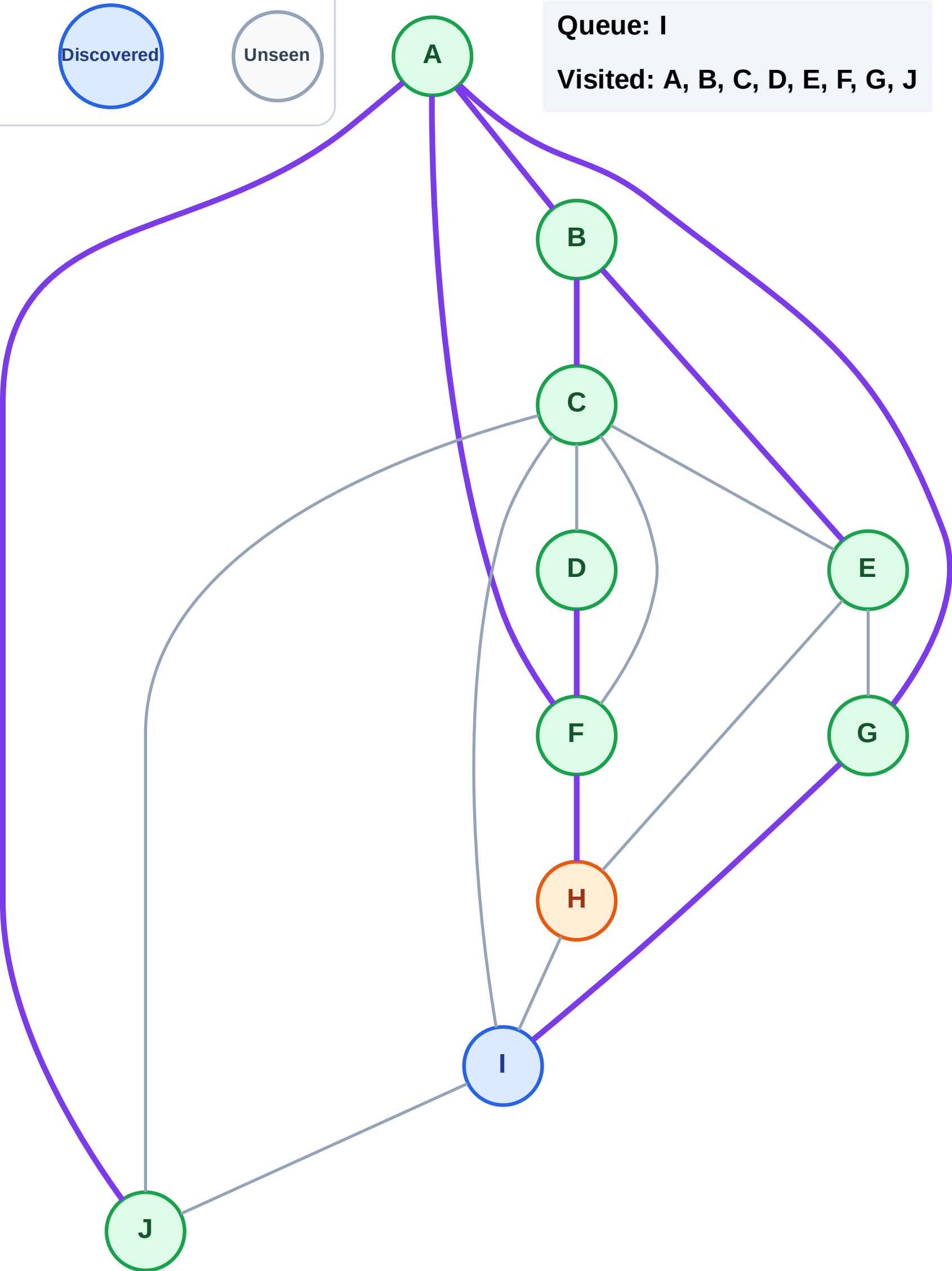
Processing

Discovered

Unseen

Queue: I

Visited: A, B, C, D, E, F, G, J



BFS Traversal

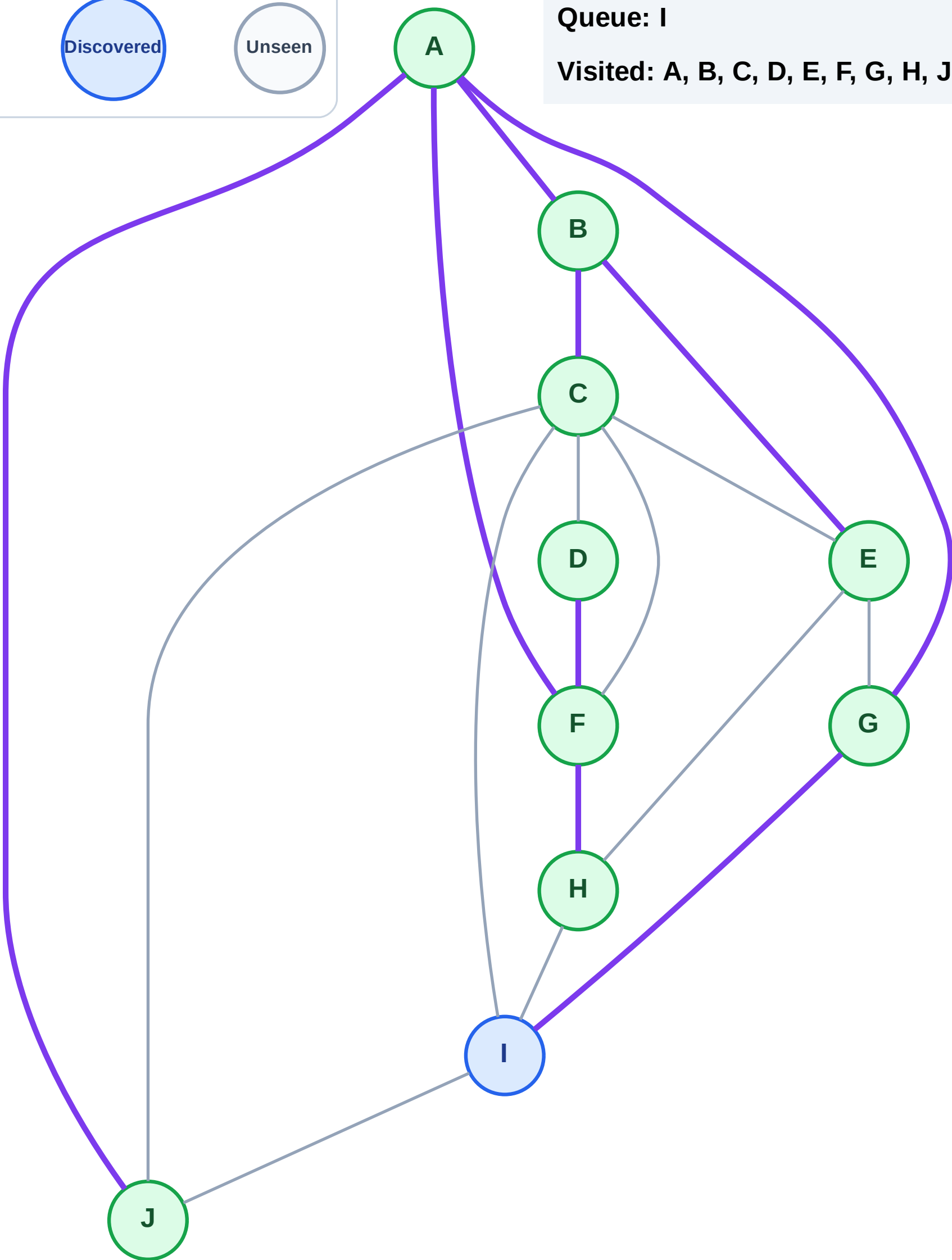
Step 19: No new neighbors from H

Legend



Queue: I

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

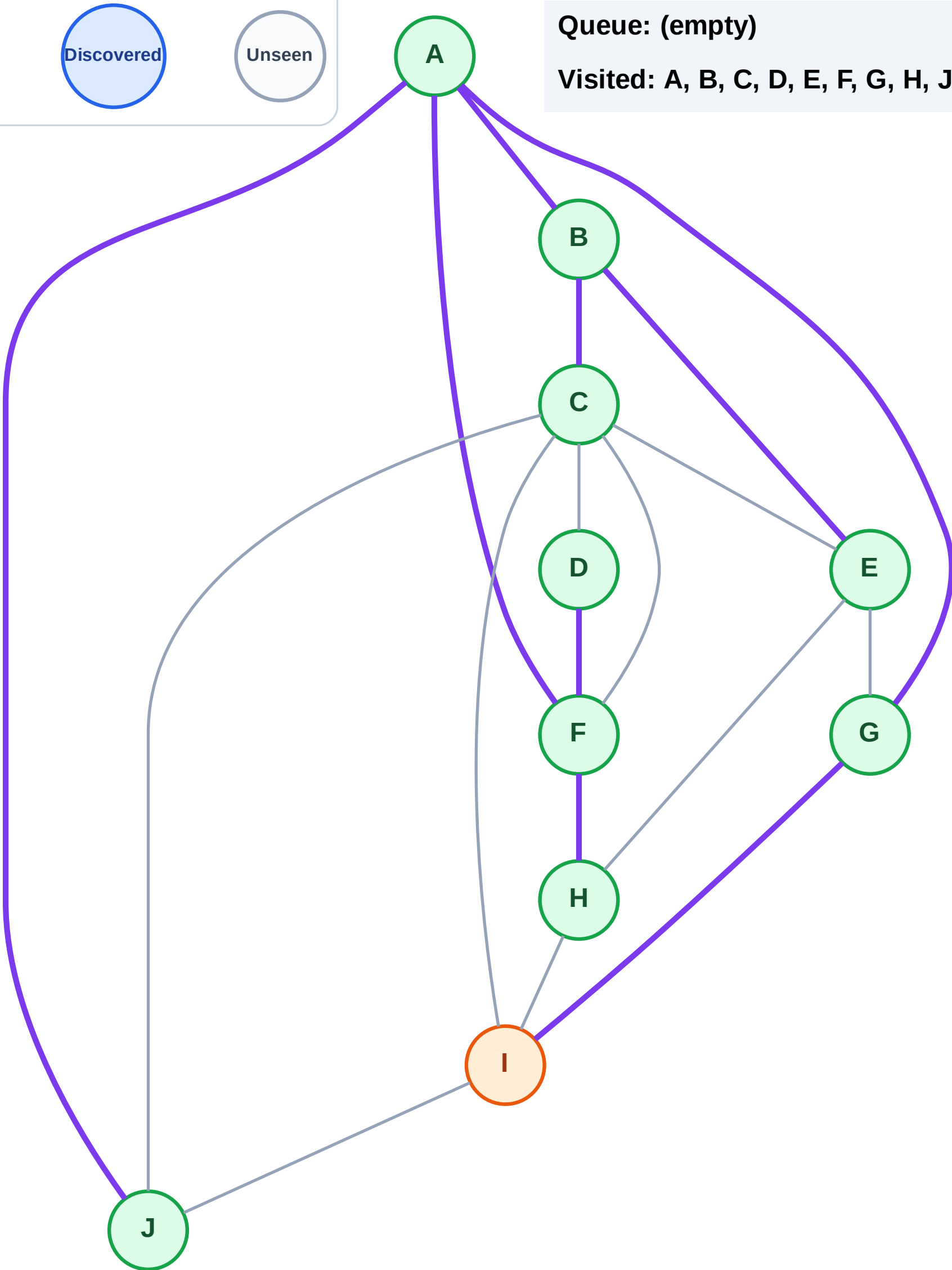
Step 20: Process node I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

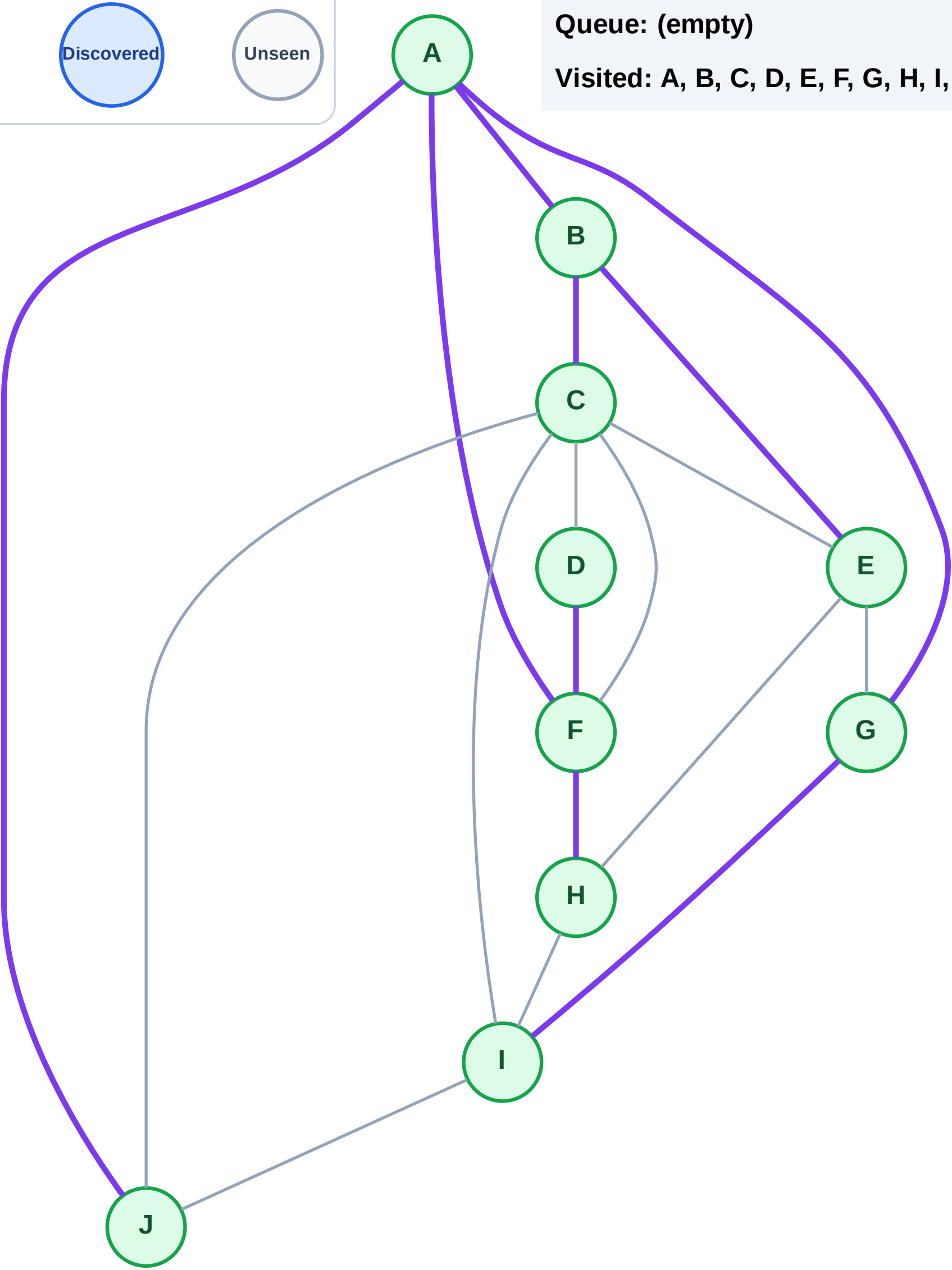
Step 21: No new neighbors from I

Legend



Queue: (empty)

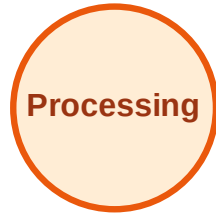
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

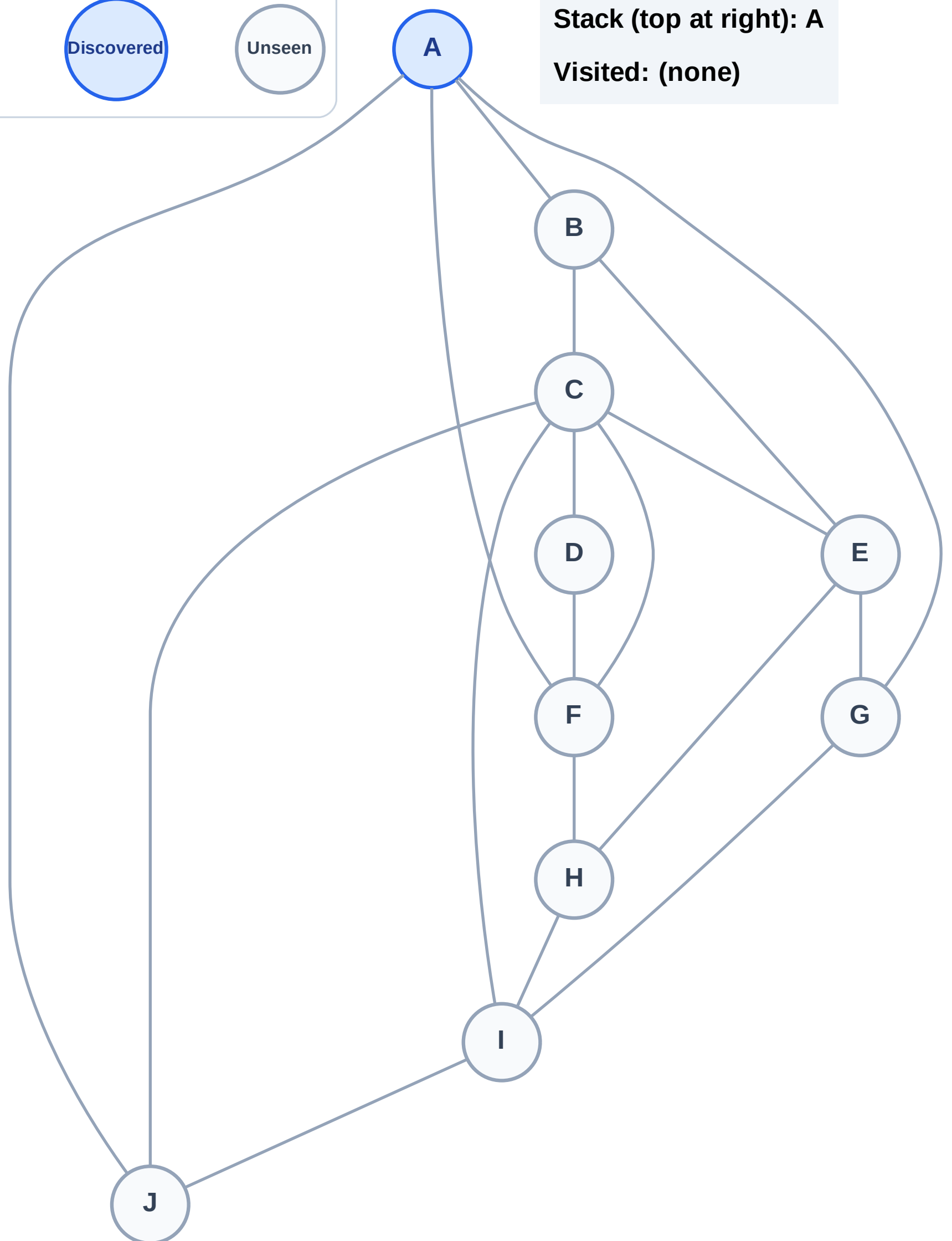
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

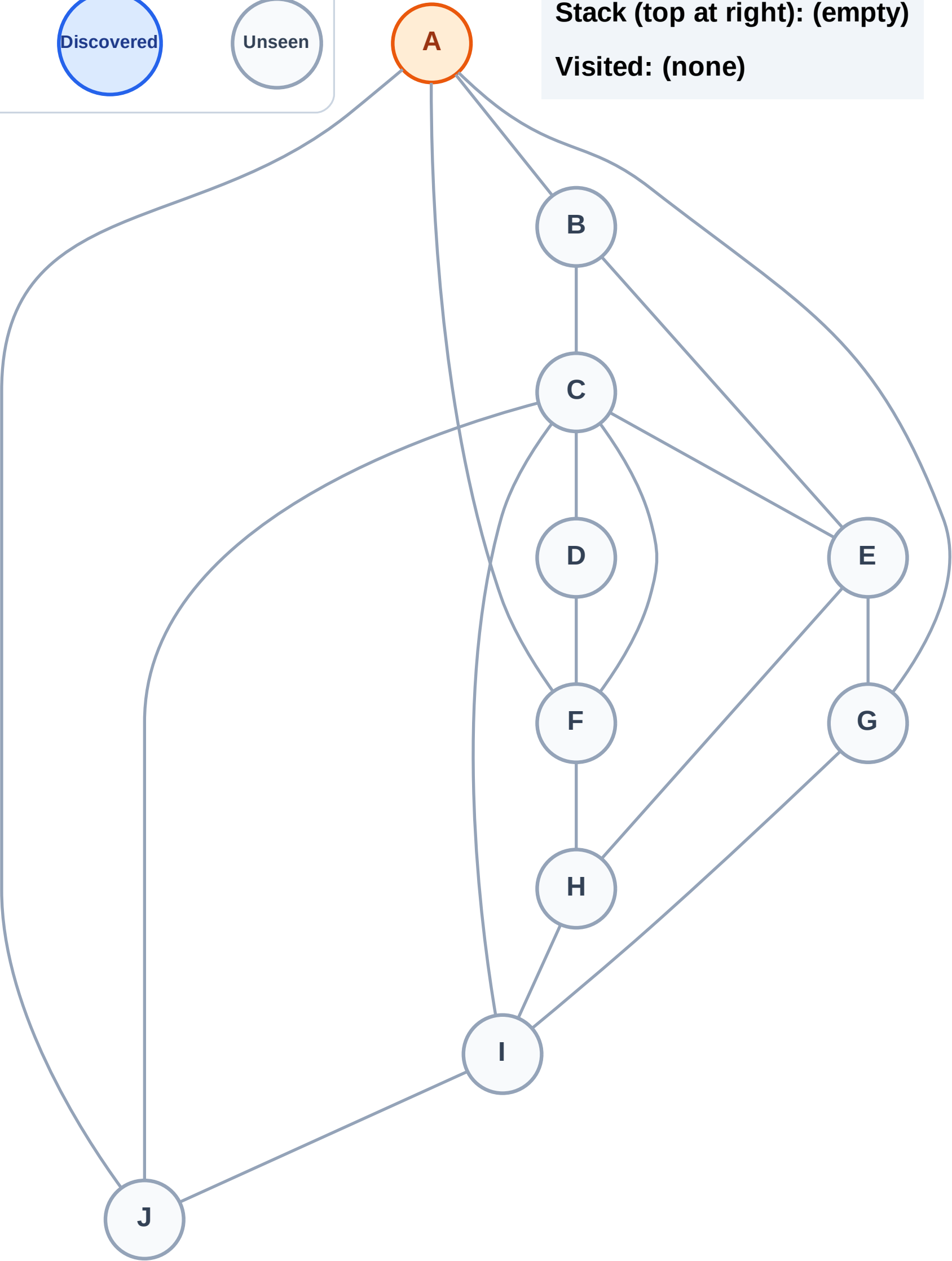
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

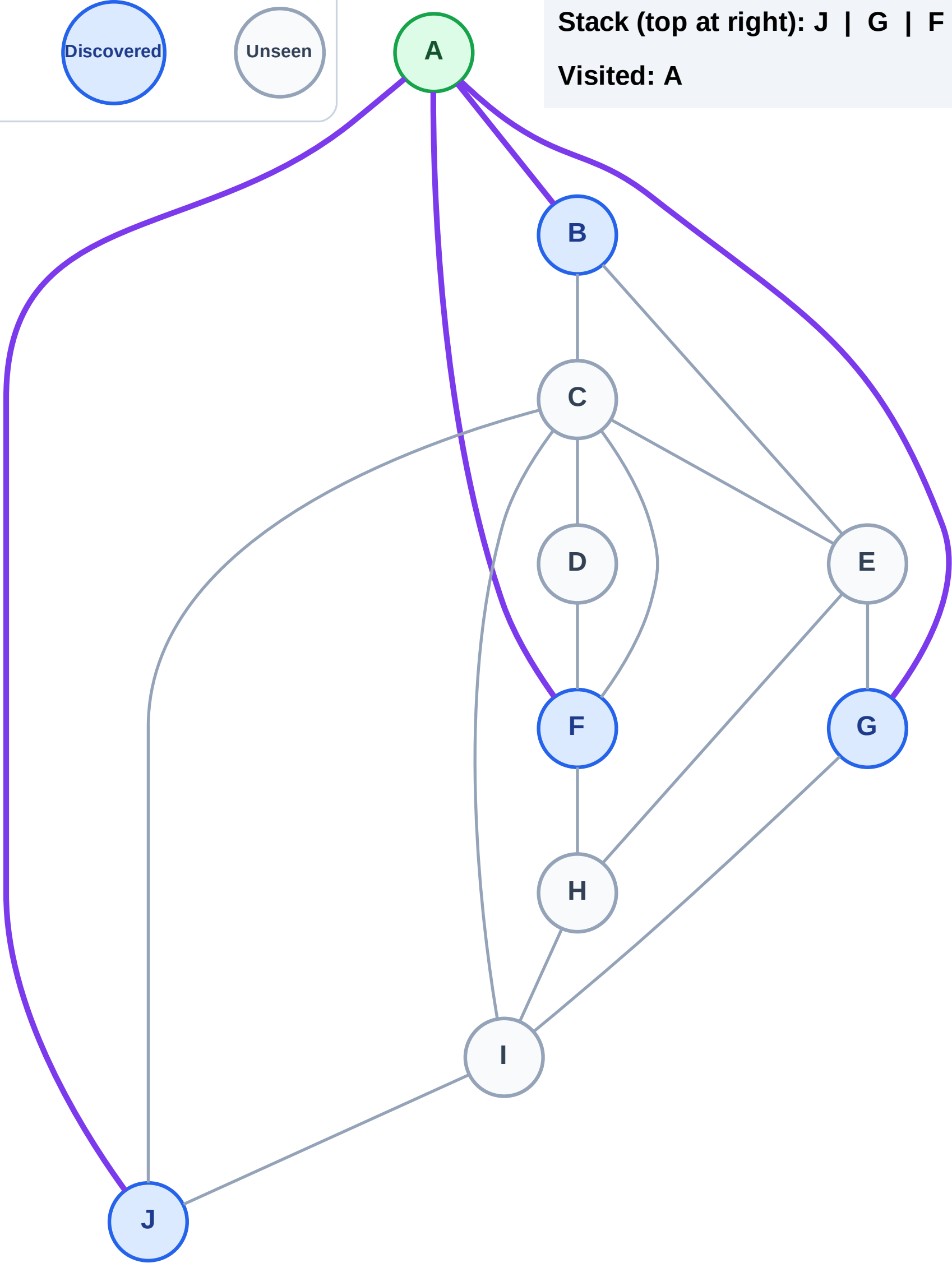
Step 3: Discover J, G, F, B from A

Legend



Stack (top at right): J | G | F | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

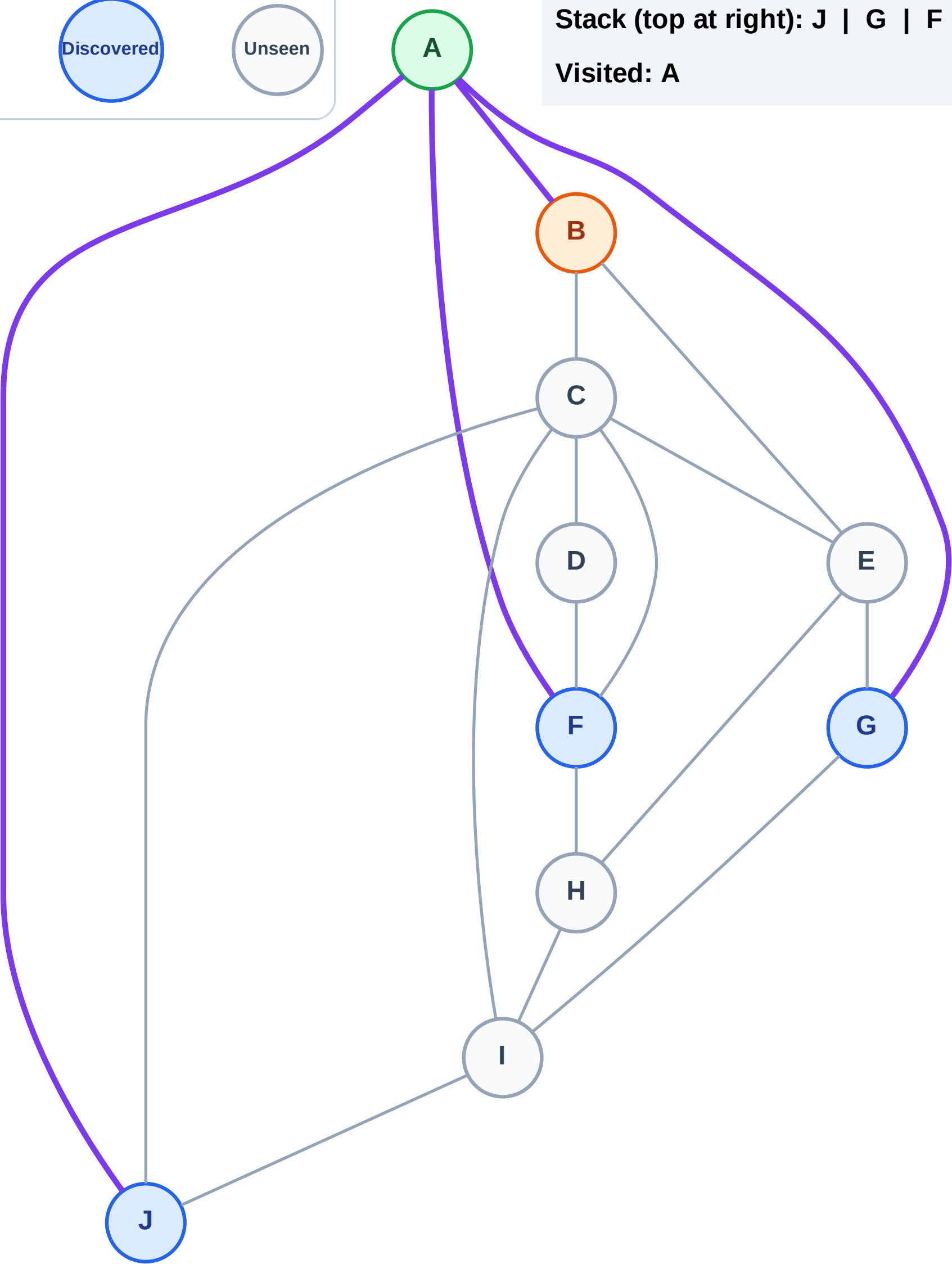
Processing

Discovered

Unseen

Stack (top at right): J | G | F

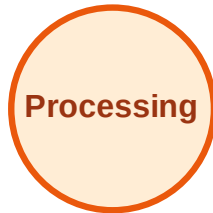
Visited: A



DFS Traversal

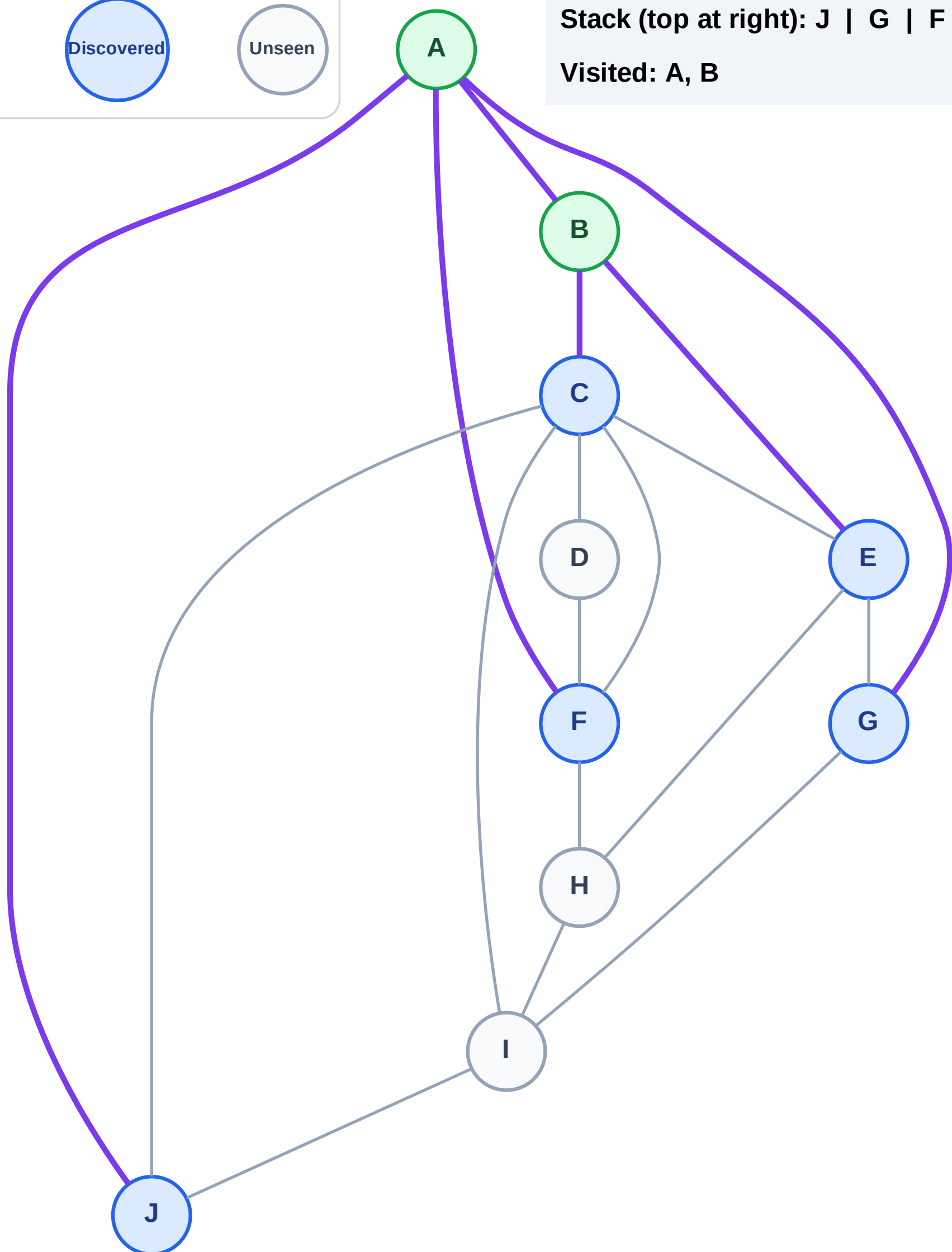
Step 5: Discover E, C from B

Legend



Stack (top at right): J | G | F | E | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

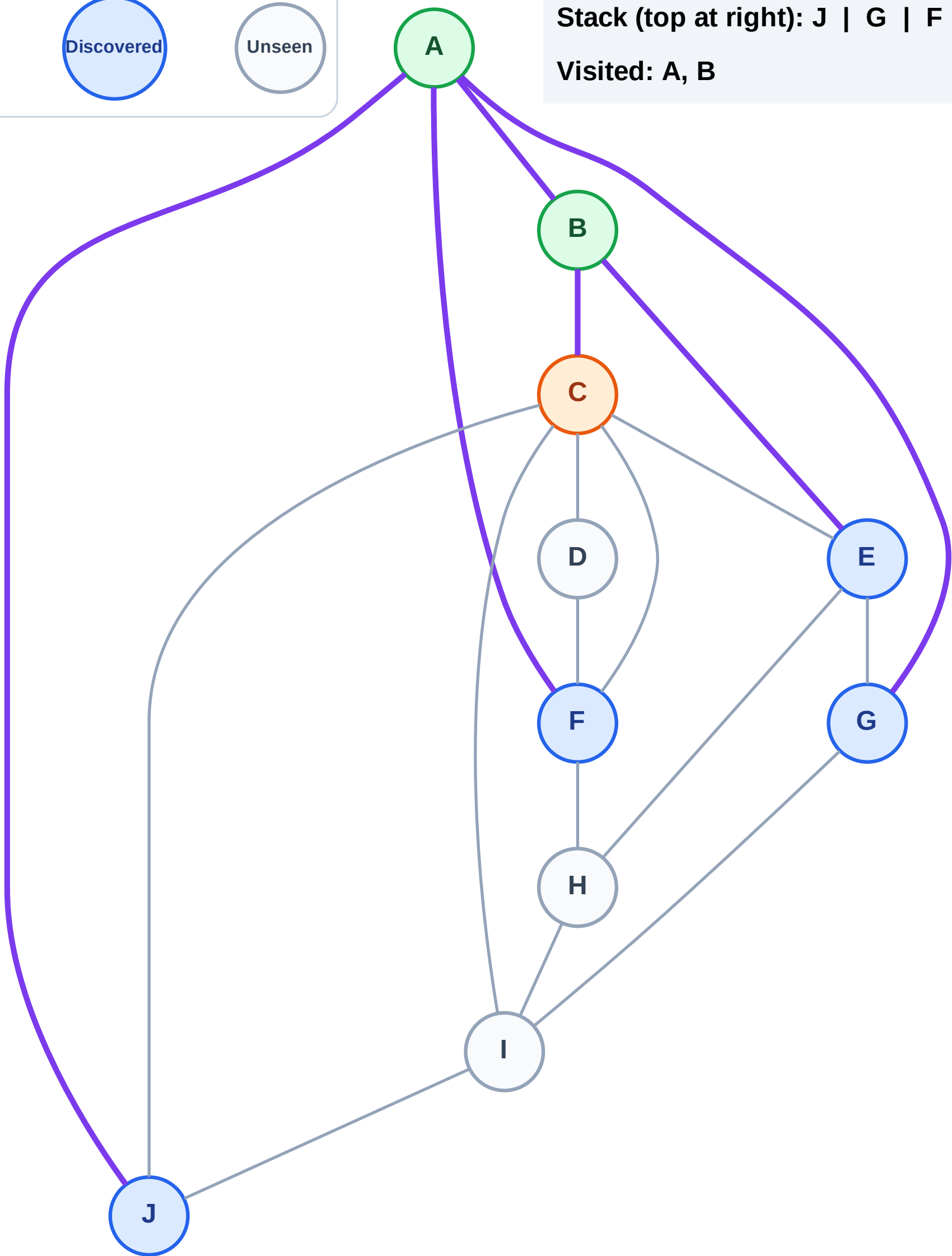
Processing

Discovered

Unseen

Stack (top at right): J | G | F | E

Visited: A, B

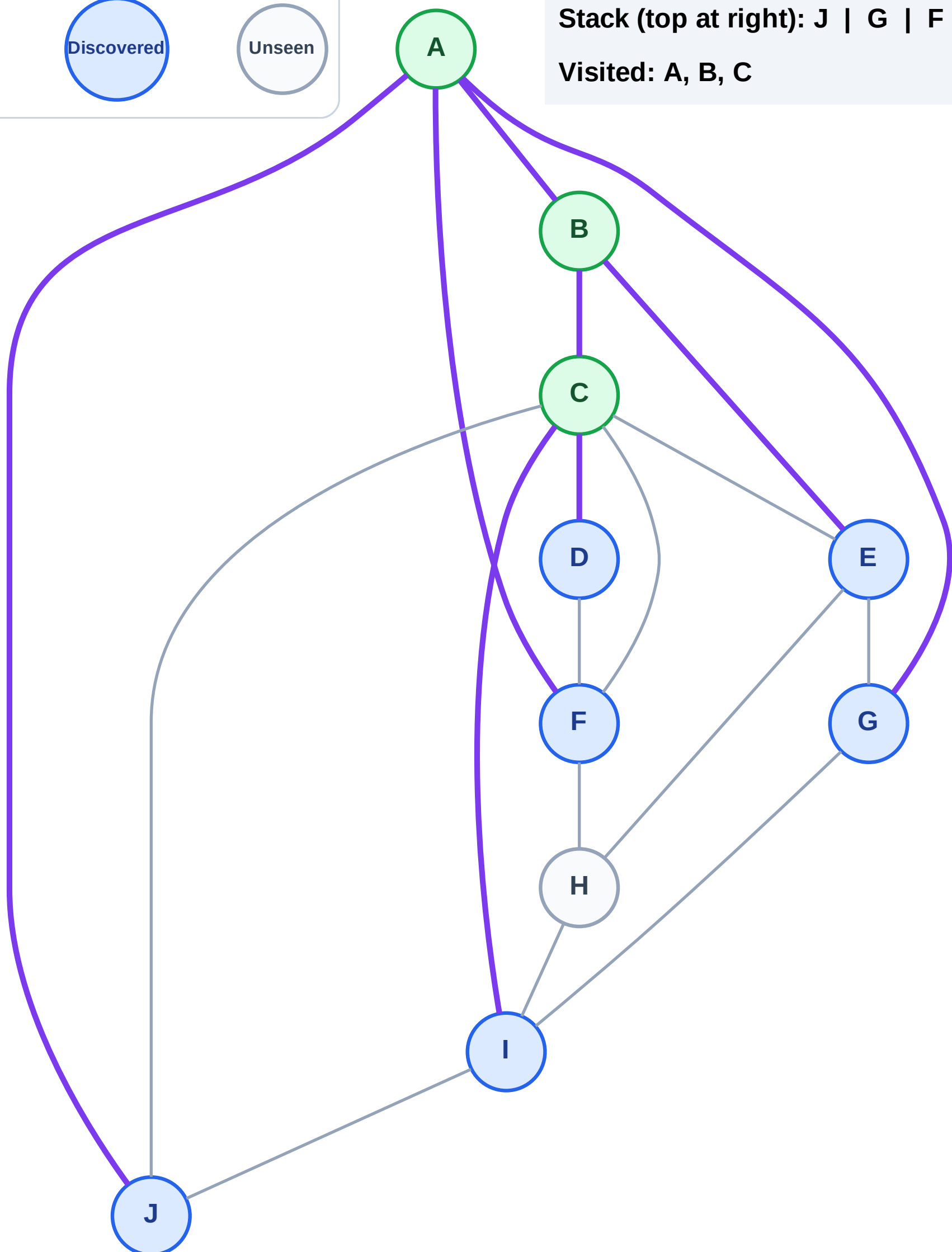


Step 7: Discover I, D from C

Legend

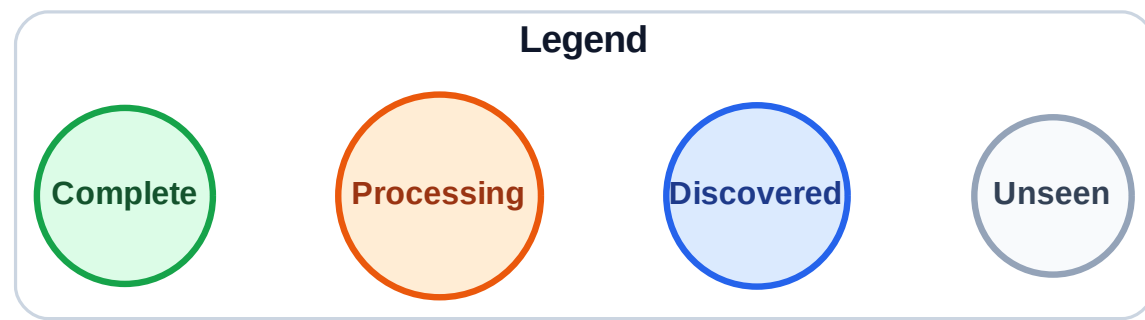
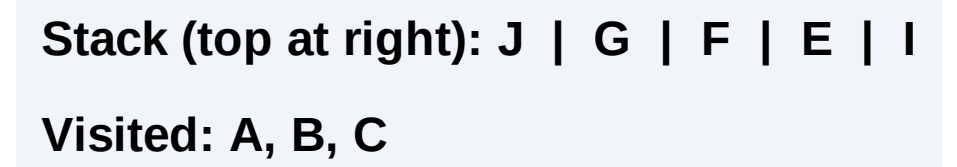
Unseen

Visited: A, B, C



Step 8: Process node D

Step 8: Process node D



DFS Traversal

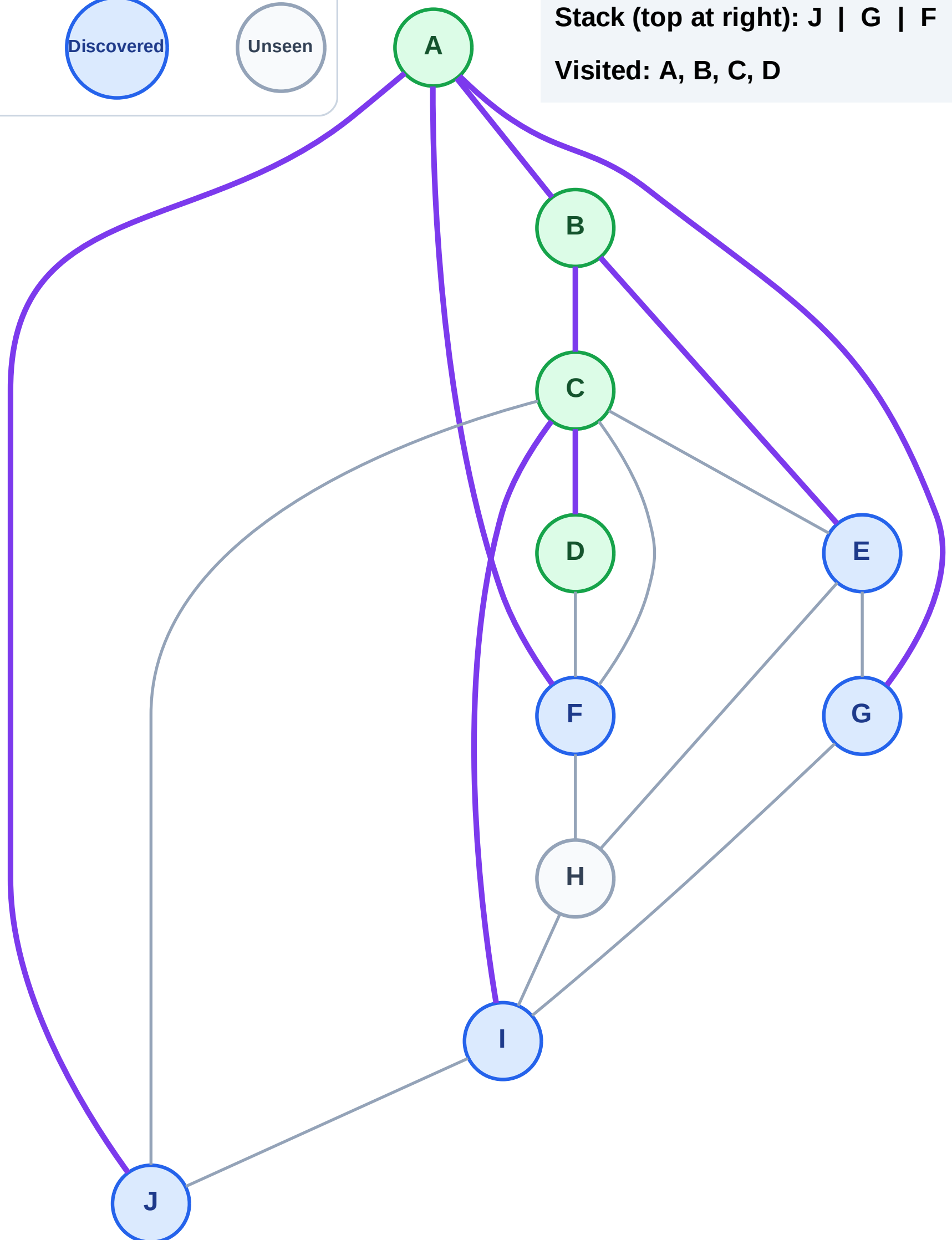
Step 9: No new neighbors from D

Legend



Stack (top at right): J | G | F | E | I

Visited: A, B, C, D



DFS Traversal

Step 10: Process node I

Legend

Complete

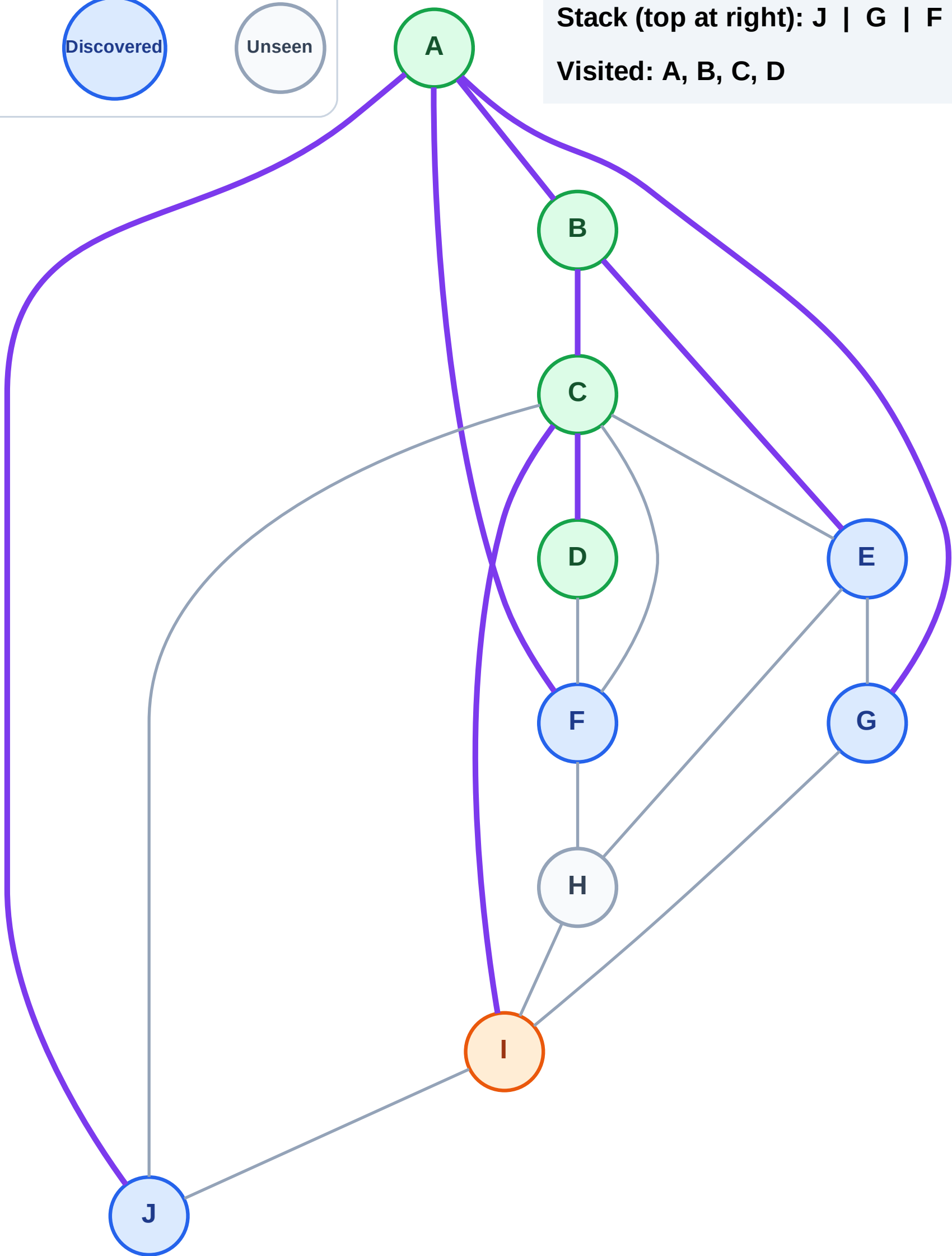
Processing

Discovered

Unseen

Stack (top at right): J | G | F | E

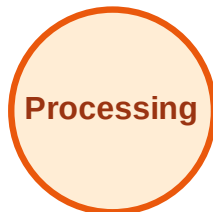
Visited: A, B, C, D



DFS Traversal

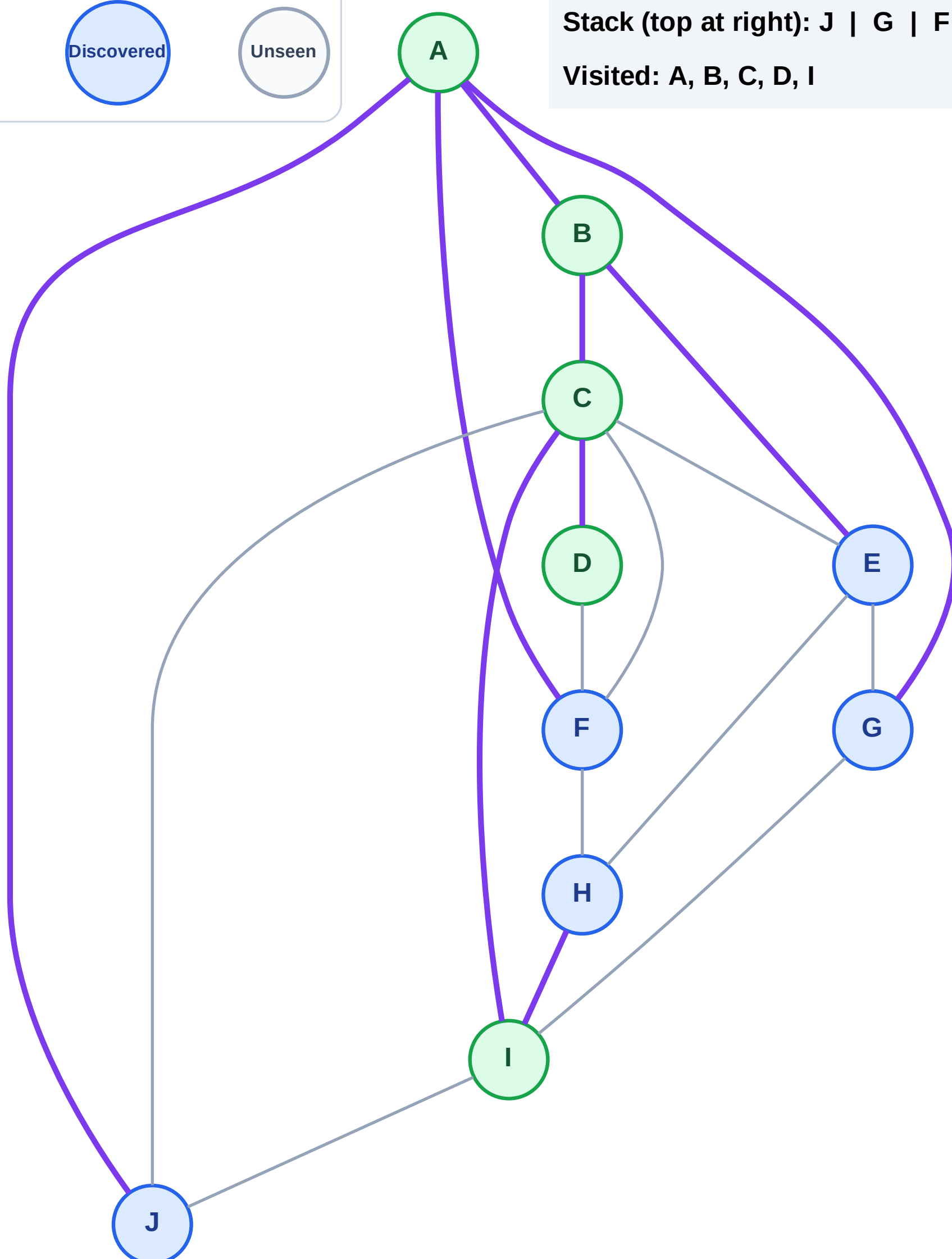
Step 11: Discover H from I

Legend



Stack (top at right): J | G | F | E | H

Visited: A, B, C, D, I



DFS Traversal

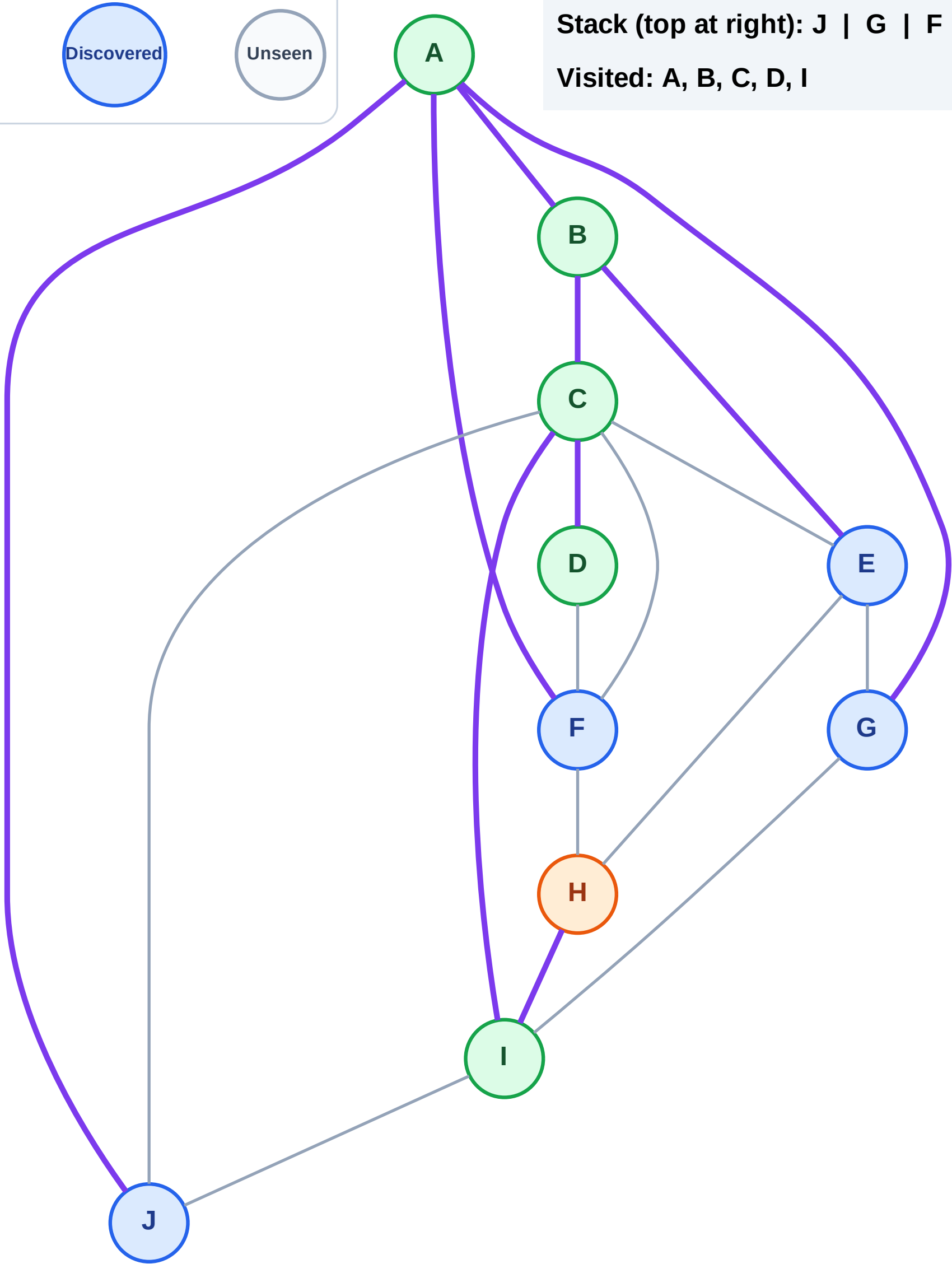
Step 12: Process node H

Legend



Stack (top at right): J | G | F | E

Visited: A, B, C, D, I



DFS Traversal

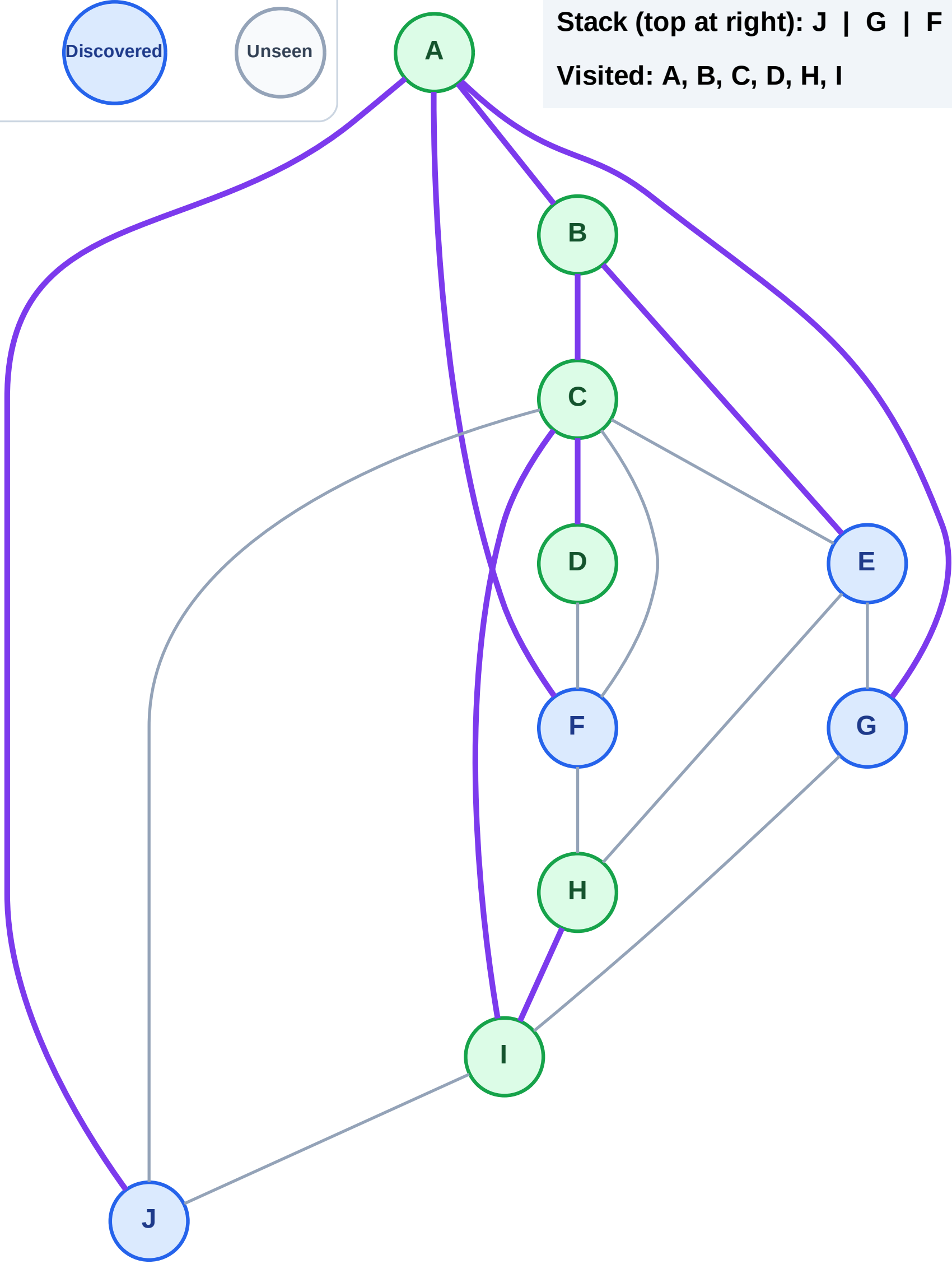
Step 13: No new neighbors from H

Legend



Stack (top at right): J | G | F | E

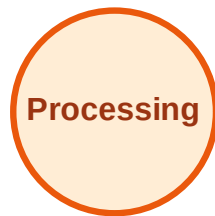
Visited: A, B, C, D, H, I



DFS Traversal

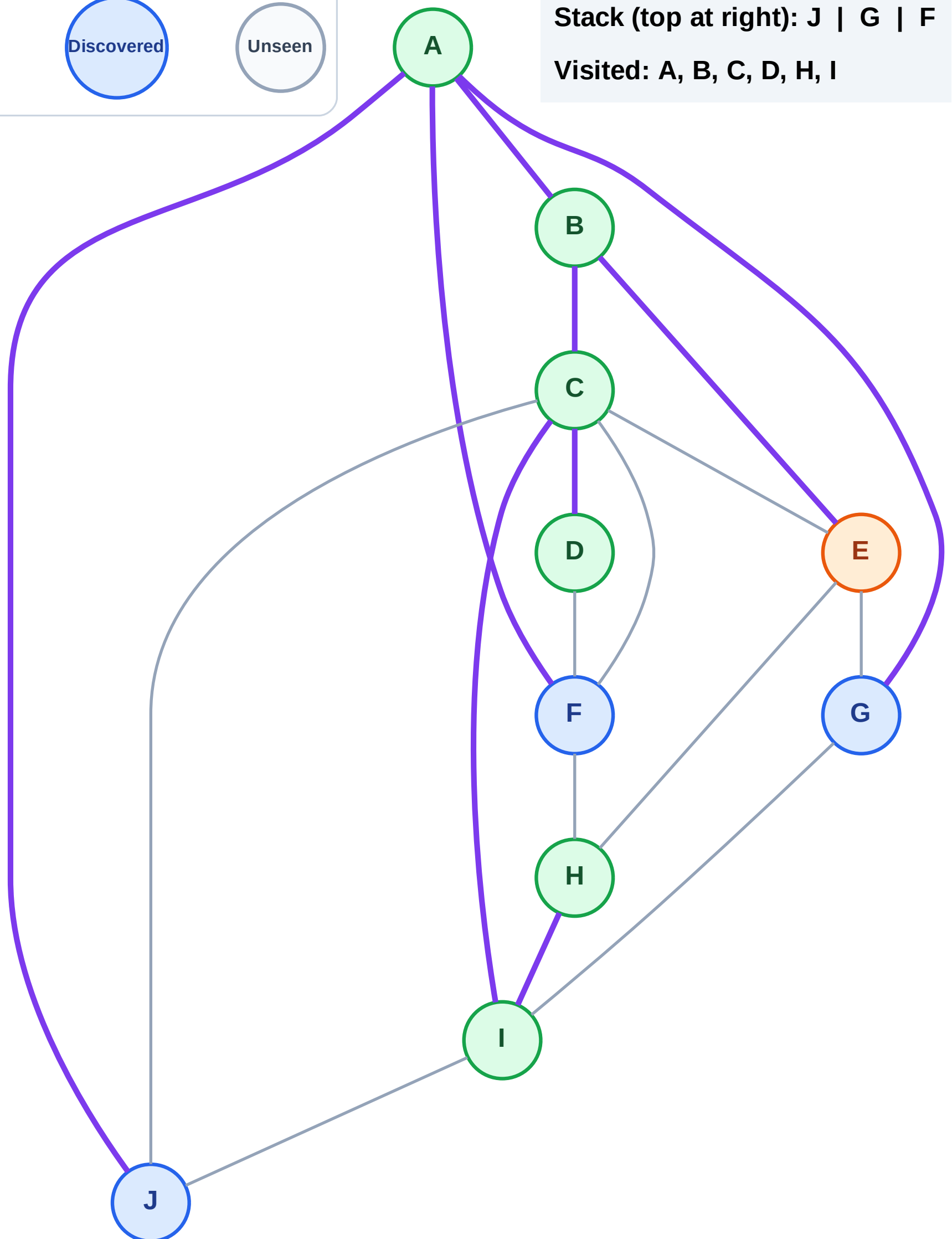
Step 14: Process node E

Legend



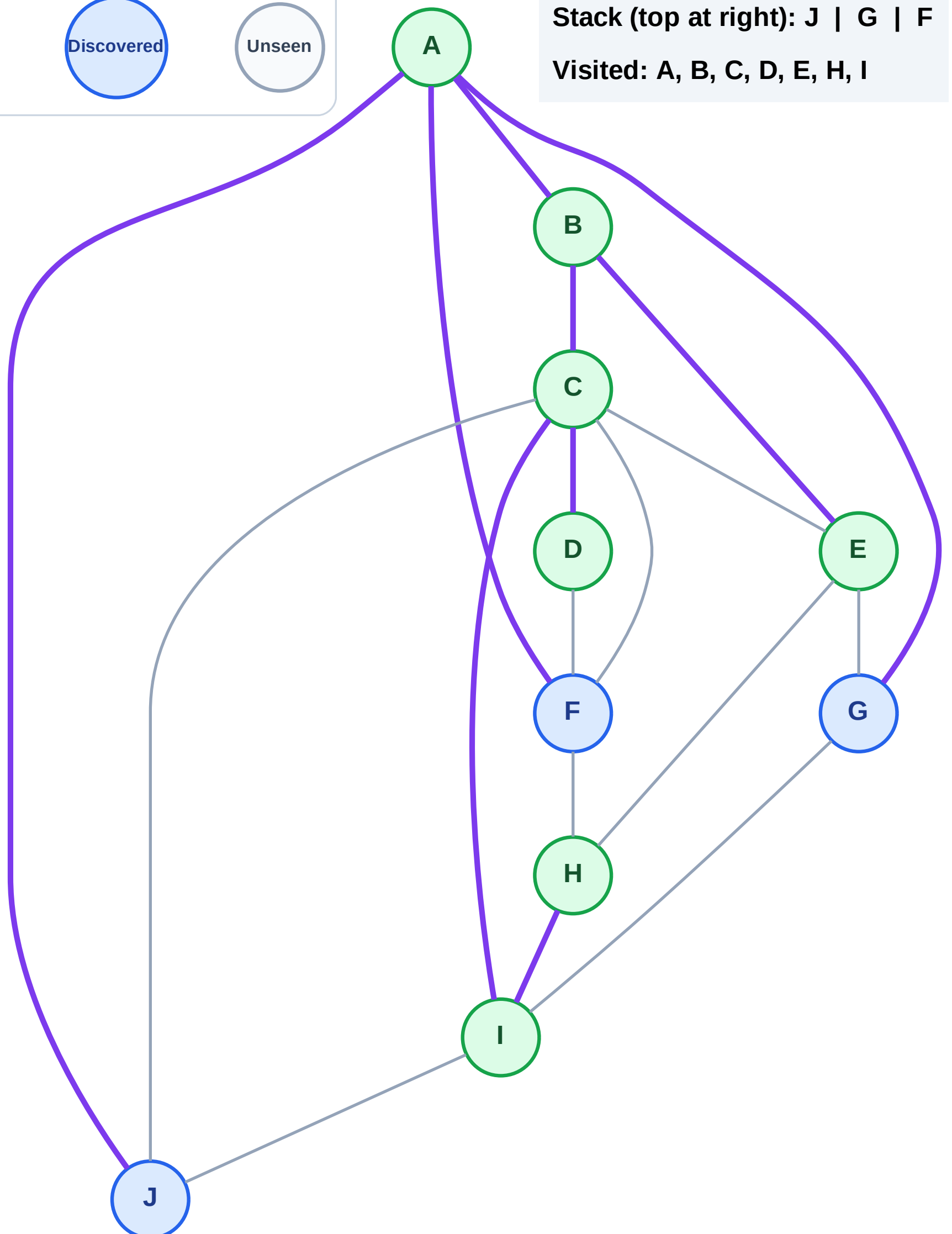
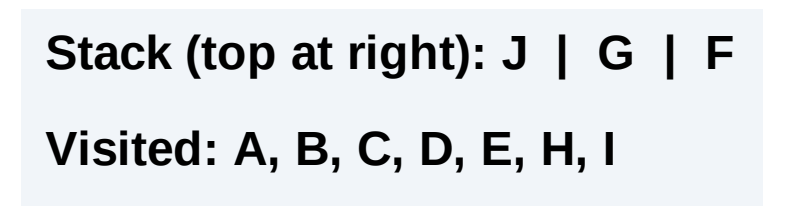
Stack (top at right): J | G | F

Visited: A, B, C, D, H, I



Step 15: No new neighbors from E

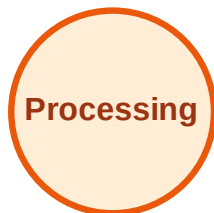
Step 15: No new neighbors from E



DFS Traversal

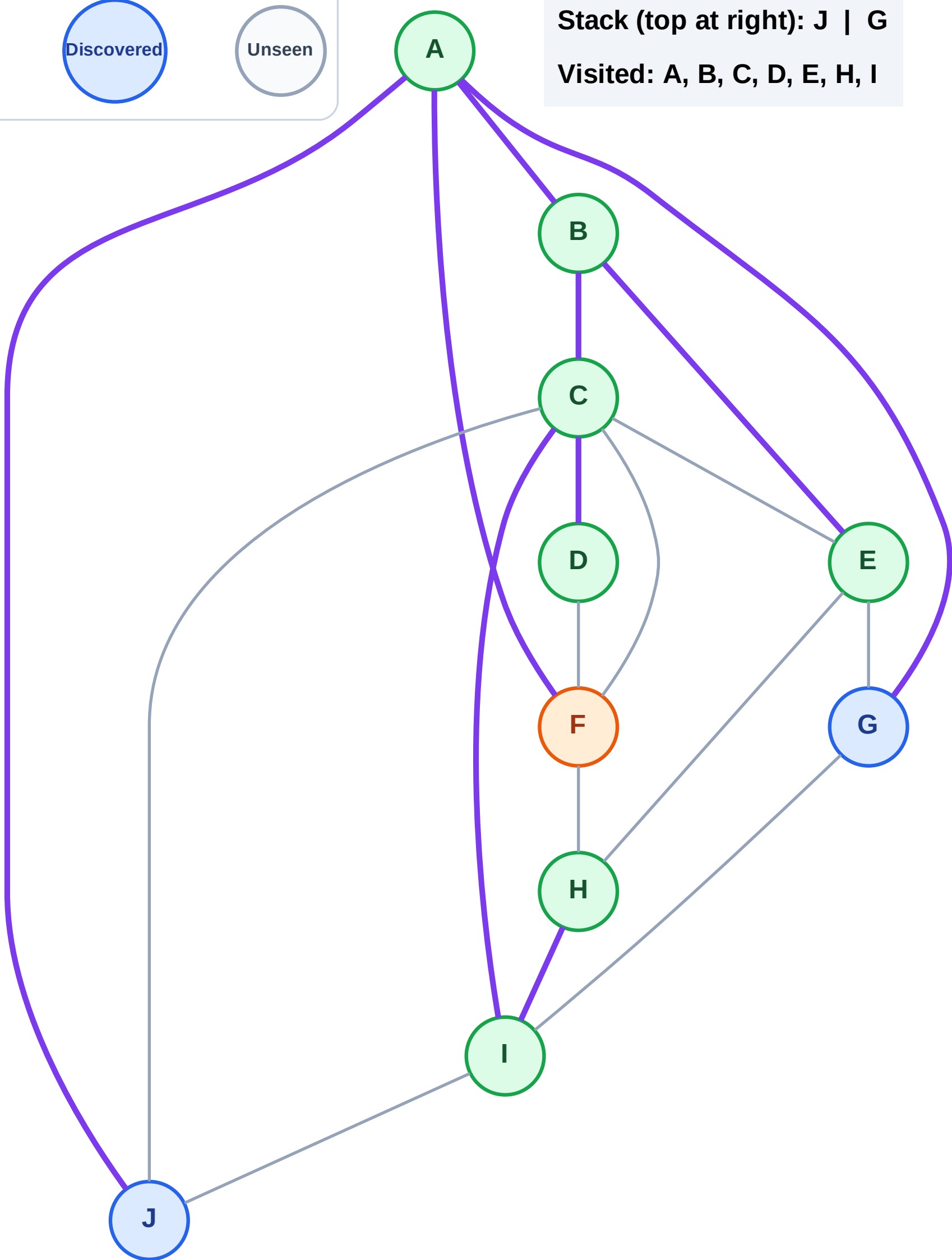
Step 16: Process node F

Legend



Stack (top at right): J | G

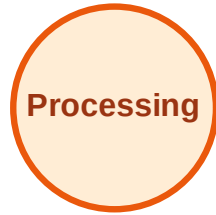
Visited: A, B, C, D, E, H, I



DFS Traversal

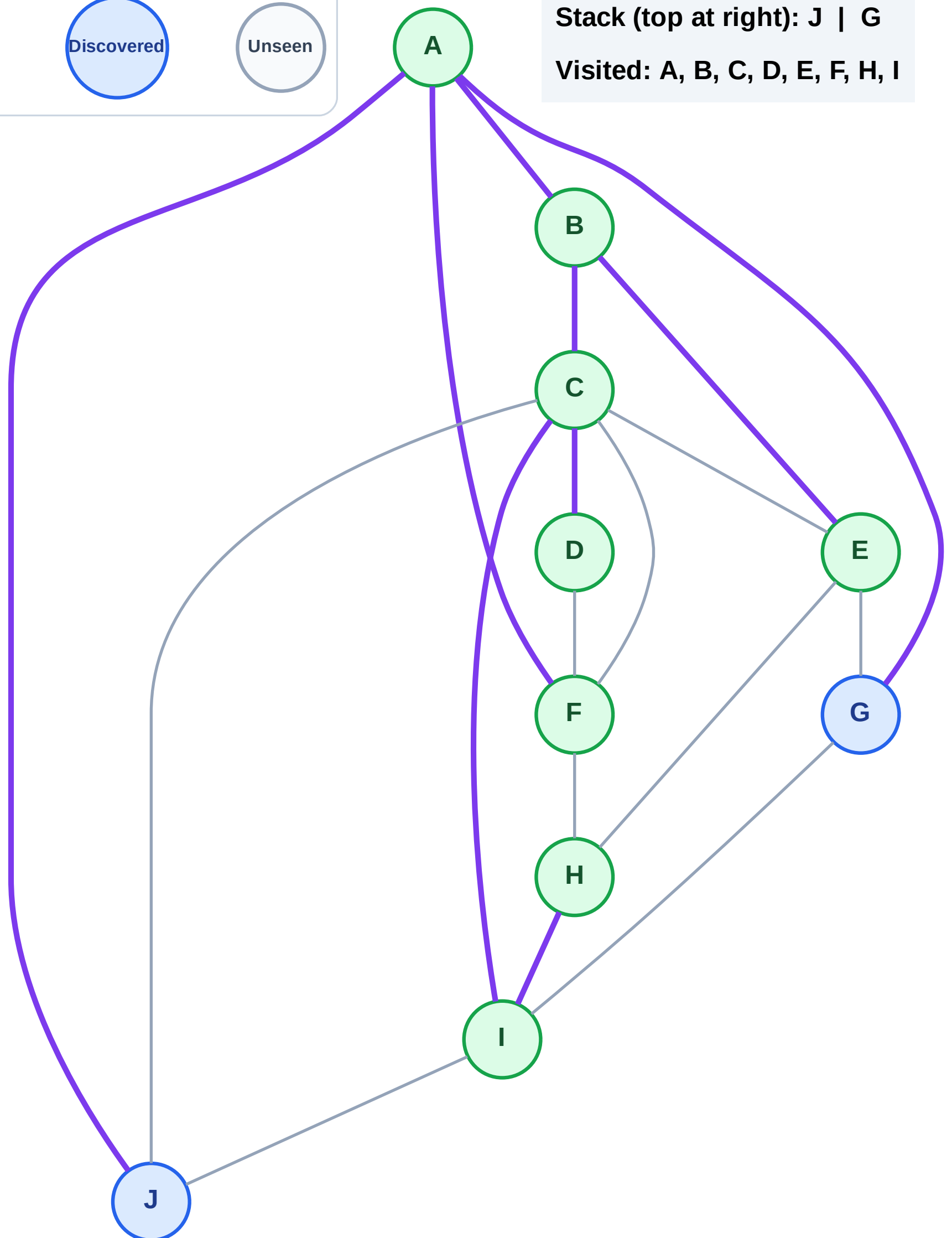
Step 17: No new neighbors from F

Legend

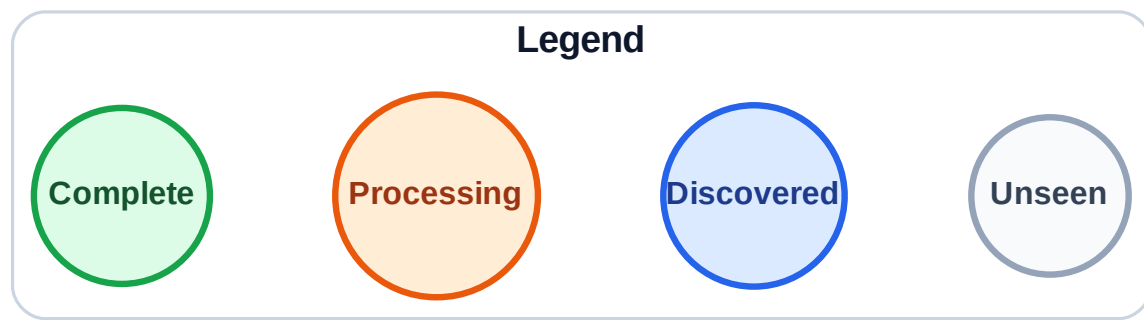


Stack (top at right): J | G

Visited: A, B, C, D, E, F, H, I

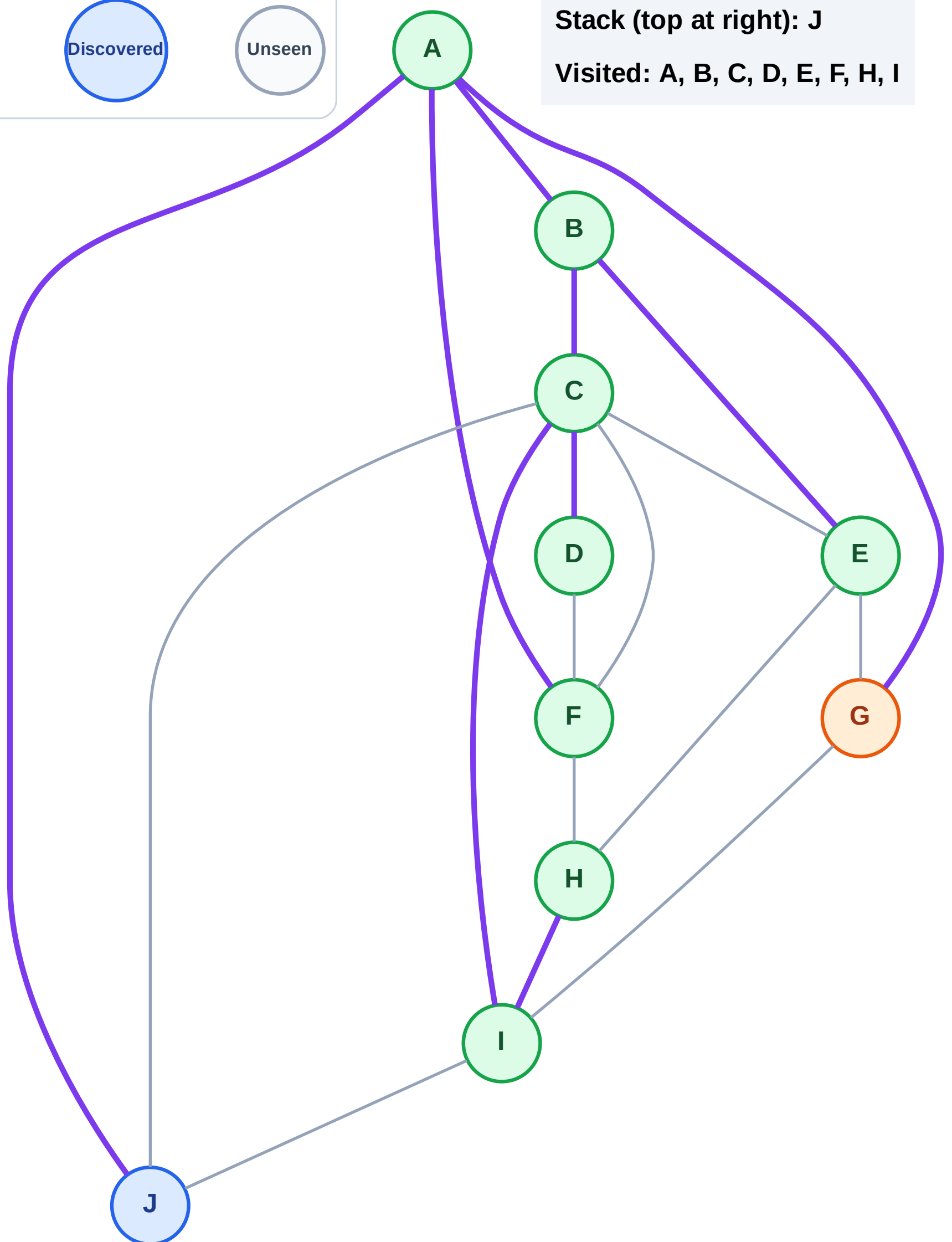


Step 18: Process node G



Stack (top at right): J

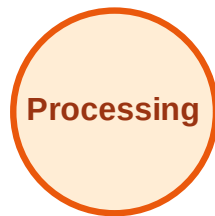
Visited: A, B, C, D, E, F, H, I



DFS Traversal

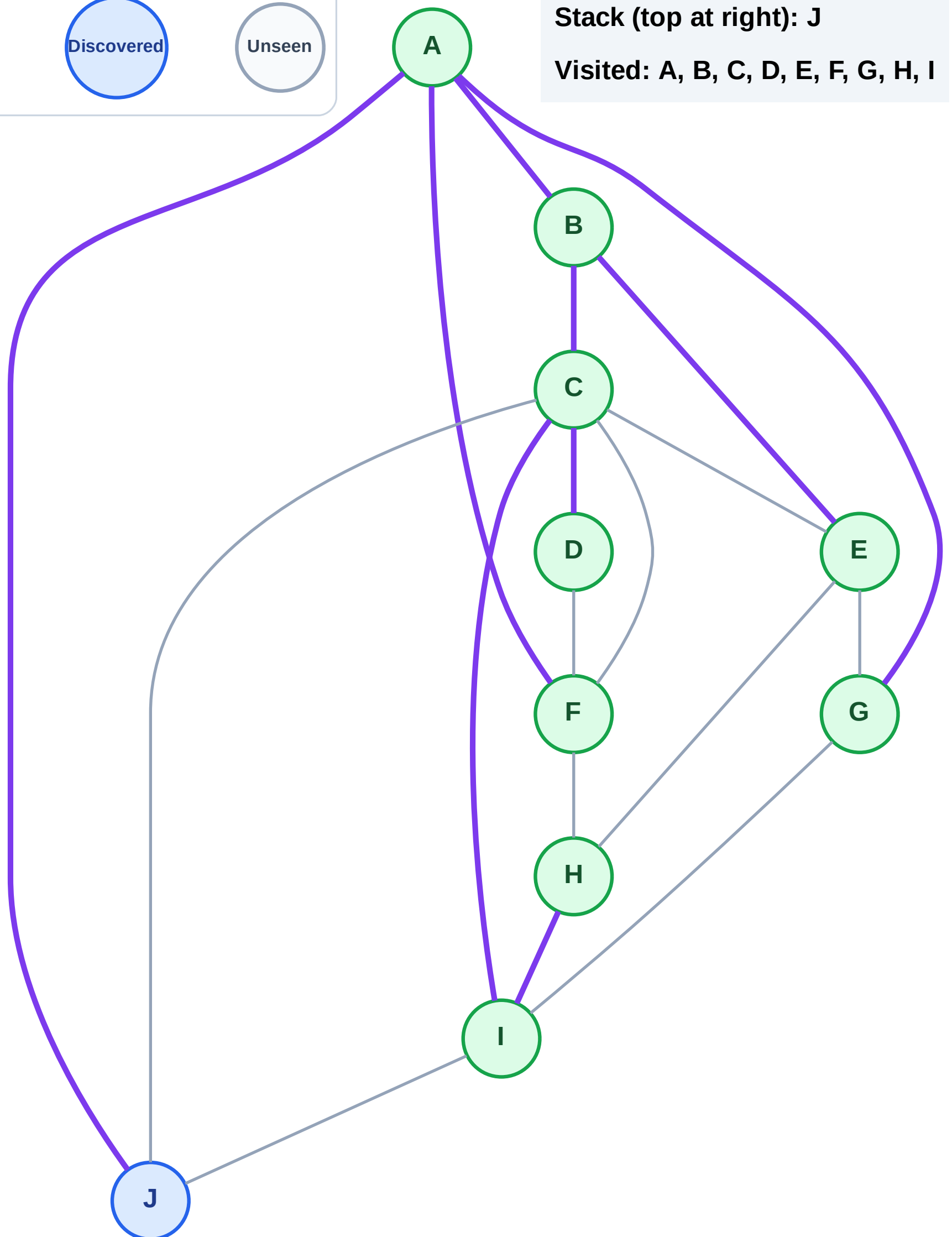
Step 19: No new neighbors from G

Legend



Stack (top at right): J

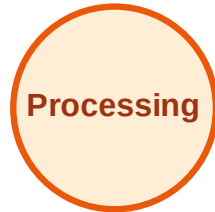
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

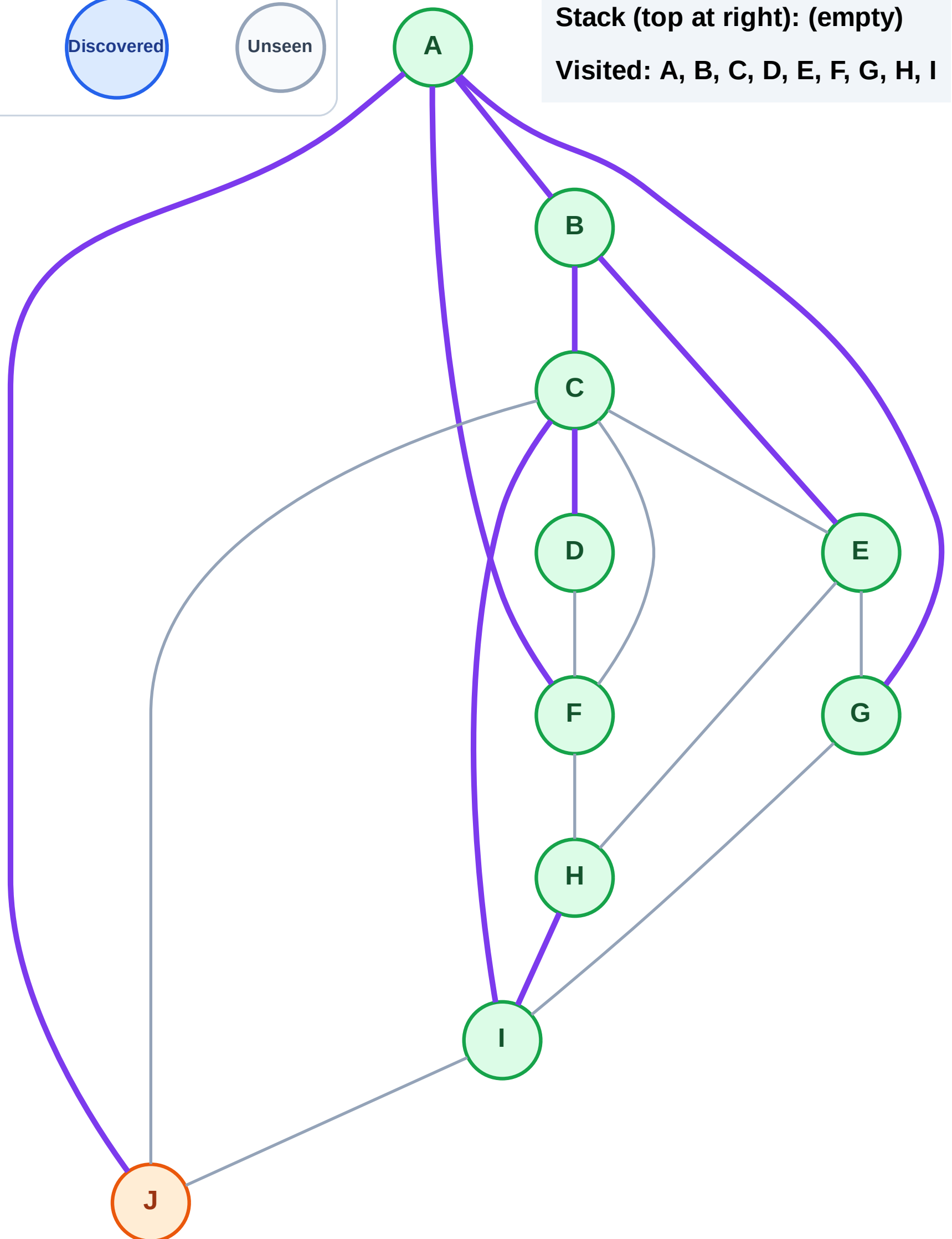
Step 20: Process node J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

